



Healthcare Industry – Provider Working Group Consultation & Recommendations

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The e-Objects concept was introduced by ACCESS Health in Theme papers in Nov 2018, which became Chapter 5 of the book 'Health Systems for New India', published by Niti Aayog in Nov 2019. Dr. Pankaj Gupta was the co-Author of Chapter 5 in the Health System for New India book. The e-Objects concept was then further detailed into Provider and Payer e-Objects along-with the data standards recommended in the National Digital Health Blueprint. ACCESS Health has now released the e-Objects design and code in opensource under MPL 2.0 license. The e-Objects concept was also discussed and recommended by the Provider Working Group and the Breakout sessions to facilitate electronic, codified data exchange between various Healthcare providers.

Abbreviations

ABPMJAY	Ayushman Bharat - Pradhan Mantri Jan Arogya Yojana (PMJAY)
API's	Active Pharmaceutical Ingredients
ASEAN	Association of Southeast Asian Nations
CAGR	Compound Annual Growth Rate
CDAC	Centre for Development of Advanced Computing
CDSCO	Central Drugs Standard Control Organization
CHI	Centre for Health Informatics
DCGI	Drug Controller General of India
DoP	Department of Pharmaceuticals
DPCO	Drug Price Control Order
EHR	Electronic Health Record
FDA	Food and Drug Administration
FDB	First Development Bank
FHIR	Fast Healthcare Interoperability Resources
HIE	Health Information Exchange
HL7	Health Level-7
HMIS	Health Management Information System
ICD	International Classification of Diseases
ICMR	Indian Council of Medical Research
ICT	Information and communications technology
IRDA	Insurance Regulatory and Development Authority
LOINC	Logical Observation Identifiers Names and Codes
MoCF	Ministry of Chemical and Fertilizers
MOH	Ministry of Health
MoHFW	Ministry of Health and Family Welfare
NDHB	National Digital Health Blueprint
NDHE	National Digital Health Ecosystem

NDHM	National Digital Health Mission
NHA	National Health Agency
NHS	National Health Stack
NIC	National Informatics Centre
PHI	Patient Health Identifier
PHR	Personal Health Record
PMJAY	Pradhan Mantri Jan Arogya Yojana
SIG	Special Interest Group
SNOMED CT	Systematized Nomenclature of Medicine-Clinical Terms
UHID	Unique Health Identifier
UIDAI	Unique Identification Authority of India

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Introduction

India's health care sector is facing a major crisis, and the poorest of the poor are bearing the brunt. In 2012, 5.5 crore helpless citizens in India were forced into poverty because of exorbitant healthcare-related expenses. Out of these, 3.5 crores fell below the poverty line due to spending on medicines alone. The ambitious National Health Policy 2017 brings within it the promise of a real and meaningful transformation in the healthcare landscape. With the launch of Ayushman Bharat, the Government of India is now committed to fast-tracking initiatives aimed at moving towards universal health coverage.

Given the scale and scope of this massive undertaking, it is a dream that can be realized only through the intervention of technology in the healthcare sector. Hence the launch of the National Digital Health Blueprint [NDHB] and the launch of the National Digital Health Mission [NDHM] - mission-mode project to implement the Blueprint and build a National Digital Health Ecosystem [NDHE]. The NDHE provides access to efficient, accessible, inclusive, affordable, timely, and safe healthcare for all the citizens.

It is perceived that India's population is digitally illiterate, the facts paint a different picture. Today, we are one of the fastest-growing online communities in the world. The spread of mobile technology and radically declining costs of data is transforming the way people connect, transact, and interact. Combine this with the grand success of Ayushman Bharat's healthcare coverage for the poor, the stage is set for the great Indian Digital Healthcare revolution.

Some interesting facts - Digital India has enabled 77.7 crores direct bank transfer beneficiaries. There are 81.32 crores mobile phone users in India and the number is still growing. With 124.3 crores Aadhaar cards issued India has covered most of its population. The UPI has crossed 10 crores users, making it the fastest adoption of any payment system anywhere in the world. To top it all, 10 crores+ PM-JAY cards have been issued.

Imagine a healthcare system driven by online Personal Health Records [PHR], where information technology is used to leverage the data. As a result, enabling access to quality healthcare to anyone, anytime, anywhere, where a National Digital Health Ecosystem nurtures collaboration, co-optation, and interoperability among all stakeholders.

National Digital Health Mission launched by PM Modi from Red Fort on 15th August 2020. ABPMJAY set the stage in 2018. India is taking the next big Digital Health to leapfrog in 2020. NDHM will serve as the digital backbone for health insurance and the provider healthcare ecosystem. The great Indian Digital Healthcare revolution is long overdue. If we can imagine it, we can make it happen.

The Three Levers of Change

There are three levers of change that can change the healthcare industry significantly:

Financial Lever:

India is moving from a pay-out-of-pocket model to a 'Universal Healthcare Coverage' model. About 60% of India's population will soon be covered as all Government Health Schemes are folding-into ABPMJAY and the missing-middle is all set to be covered too. Total claims to be adjudicated are set to go up by 10-13 times. Digital processing is the only way to manage the upcoming tsunami of claims.

It is in the interest of the provider sector to comply with the change, as the insurance providers will soon start asking for eClaims in a standard format supported by eDischarge and eEncounter eObjects for a faster turnaround on claims. *Details of these eObjects are provided in Appendix: eEncounter eObject, ePrescription eObject, eClaim eObject, and eDischarge eObject.*

Epidemiological Lever:

Data Science holds promise to community medicine interventions through the analysis of high volume and high variety of data, using high-velocity feedback mechanisms. Here, high variety - refers to the data collected from disparate sources, such as combining data from electronic medical records with purchase histories or social media profile updates. High volume - refers to data with orders of magnitude more observations and/or variables per observation than prior datasets in the domain. High velocity - refers to a data generation process wherein data are compiled and analysed in real-time or nearly in real-time, often by algorithms operating without human intervention.

World over, Health Insurance does not just pay for secondary and tertiary care, it also absorbs, or offset healthcare costs associated with a wide range of services, such as routine health examinations, unforeseen illnesses or injuries, inpatient and outpatient surgery, specialist referral visits, and prescription medications. etc; it is in the interest of Health Insurance providers to reduce the overall disease burden. Digital will help us catch the disease earlier at primary care levels, and it will manage the referral network better. Data-driven clinical protocols will help slow down the disease progression. Evidence-based medicine is needed to reduce the disease burden. Some indicative examples - John Hancock shifts from Life Insurance to Disease management; CVS purchased Aetna; UnitedHealth bought a large Doctors Group. ICICI Prudential Life Insurance Covers Critical illnesses.

The Covid19 pandemic did not break the healthcare system, but it just exposed a broken system. Today Public Health managers cannot perform resource optimization because they do not know how many Doctors, Nurses, Beds, Hospitals, Assets exist or are readily available. It is certainly not going to be the last epidemic hitting us. In this pandemic, and every next time around we need to be better prepared with

resource optimization tools, predictive analytics and armed with current epidemiological studies to tackle it. We need to have information available to decide what interventions are needed and how to carry out program planning, revision, and evaluation.

e-Governance Lever:

‘One India - One Health’ portability refers to the facility provided to the beneficiary such that the beneficiary can avail the healthcare services and claim the due benefits anywhere in the ecosystem across Districts, States, UTs, and the Nation.

Telemedicine is a tool that can be applied to a variety of business models, for example -. Telemedicine can be used for Virtual Consults in acute illness or communicable diseases nearly at par with physical in-office consults. Furthermore, telemedicine can also be applied for disease management and regular follow-ups in chronic or non-communicable diseases.

Availability of the prescribed and refill drugs through the ePharmacy is a very strong lever for the adoption of the digital systems by the beneficiaries because drugs form a big part of every treatment.

In times of the Pandemic, the vaccines delivery has assumed enormous significance and challenges in the entire Ecosystem. A Digital backbone makes it easier to identify, trace, track and vaccinate the hot spots based on the population. The Digital backbone combined with the DigiLocker will be useful for mission Indradhanush and regular vaccinations too.

Diversity of India’s Healthcare Provider Sector

The provider sector in India is unlike anywhere else, due to its unique characteristics.

India has a remarkably diverse population where the language, food, and culture change every 50 miles. India officially has 29 Languages. The provider sector is characterized by tremendous inherent diversity. It is impossible to superimpose a foreign system on such diversity. The system must respect and adopt diversity and not vice versa. Hence a home-grown system is essential for India.

- Lopsided distribution of Doctors
 - 9,00,000 Doctors in India
 - 50% in 4 large metros
 - 40% in Tier 1 and Tier 2 cities
 - 10% in semi-urban and rural India
 - There are Only 25,000 Seats in medical colleges for specialist doctors every year.
 - Most doctors go into primary and secondary care.

- Women and Child Health remain the focus because India adds the equivalent of 1 Australia to its population every year.
- Facilities: 90% have below 50 beds.
 - India has 1 Bed to 1,000 Population.
 - WHO recommends 1 Bed to 300 Population.
 - 5% are > 150-bed facilities.
 - 5% are 50-150 beds facilities concentrated in Tier 1 and Tier 2 cities.
 - 90% of beds are in small Clinics and medium-sized Nursing Homes < 50 Beds.
- A large number of diagnostic labs, but very few are accredited.
 - Over 1,00,000 labs
 - There are only about 2,000 - 6,000 qualified doctors with pathology qualification, that means most of the labs operate without a pathologist or a doctor.
 - There is a severe shortage of radiologists. Only about 10,000 - 12,000 practising radiologists. The radiologists to population ratio are 1:1,00,000.
 - 60% are unorganised small and medium size Labs.
 - <20% are organised labs.
 - 20% are hospital-based labs.
 - Only 1% of the labs are accredited by NABL.
 - Insufficient regulatory framework for radiology
 - Total market size is between INR 40,000-60,000 Cr.
- A large number of pharmacies - disproportionate to the number of pharmacists
 - 65,000 stockists and 75,000 to 10,00,000 retail pharmacies
 - There are about 5,60,000 registered pharmacists.
 - 6% of the total sales come from organised pharmacies
 - 6% of the total sales come from hospital pharmacies
 - The largest selling brand of the drug in any common therapeutic category has no more than 8%-10% market share and that too in pockets at best. Not only that, the same brand name, could be drugs of completely different therapeutic classes in different regions. This is hazardous.
- Top Health Information Management System products have quit the market, only digital start-ups are left.
 - **Tier I** – iSoft and TrakCare have quit the market.
 - **Tier – II** – TCS, Wipro, Religare – shut shop.
 - **Tier – III** –Struggling with Client-Server era technology. Some like Akhil, Srishti – have transitioned to web technologies.
 - **Digital Start-Ups** – Hold a promise to take over the middle and bottom layers of the pyramid. They can adopt the latest tech. They are well funded but lack size and scale.

- **Workflows, Terminology, and Rules** are usually a standard in every Industry vertical but in Healthcare all three are not a standard. This gets amplified with the regional nature of India where every Region and the State of India is diverse and fiercely protective of its workflow, language, and culture like an independent nation. This has profound implications on the interoperability of systems in the larger context. If Uttar Pradesh [UP] was considered a country on its own, then from a population perspective UP would be the 5th largest country in the world. UP is home to 20 crores people but is less than 8% of the overall land area of India, hence one can imagine the diversity that lives in the whole of India.
- **Metadata Diversity** the diversity discussed in the preceding points leads to a very significant implementation challenge. India has Metadata very different from the rest of the world e.g., Lab Panels, Drug Brands, Combo Drugs, Specialty Names, Facility types, Department Names, etc. India cannot be superimposed with profiles and code sets from outside India, those are alien to it. We strongly recommend using opensource standards, profiles, code sets customized for India, consistent with the spirit of #AtmaNirbharBharat #VocalForLocal.



Red Flag: Push Marketing against Indian Healthcare Culture

- Do not Jump - Learn to drive on an Alto before a Porsche!
- International profiles and Code sets
- Foreign Metadata



Green Flag: Respect the Culture – Change will sell itself.

- MeitY Open Standards
 - India specific Metadata and Data Dictionaries
 - Minimum Viable Products - Microservices
 - eObjects for Interoperability
 - Start-ups in the Social Entrepreneurship Accelerator are pulling.
- Health Insurance Coverage has been small and Pay-out-of-pocket has ruled the market and hence, the diversity. Only now with the introduction of PMJAY, we are likely to cross the 50% mark from a Health Insurance perspective. Thus, Health Insurance now comes into a position of strength for making the provider side behaviour conform to the digital health standards in the larger national interest. However, Health Insurance will have to respect the local culture and diversity to achieve any compliance on standard profiles, code sets, and metadata.
 - Private and Group Health Insurance 3%
 - Other Government Schemes -10%
 - PMJAY – 40%
 - Top Tier, Uninsured, Missing Middle - 47%

Fuzzy Boundaries

Digital Health inherits the fuzzy boundaries that pre-exist in the healthcare sector. These fuzzy boundaries get accentuated and cause a lot of disruption when applying digital health to the healthcare sector. Listed below are some instances of the fuzzy boundaries existing in the healthcare sector:

- **Healthcare is a State subject.** The delivery of healthcare largely rests with the States as health is defined as a State subject in Schedule 7. On the other hand, the Central government funds 85% of most public health programs. However, all the regulation and execution authority vests with the States. The states are free to accept or reject any healthcare initiatives or guidelines from the center. India is too diverse to expect a single healthcare solution across all States. Unity in this diversity is elusive in this context.
- **MoHFW regulates public health only.** Most of that is done by respective directorates under the various Disease Control Programs. The rest of the Healthcare eludes MoHFW.
- **Health Insurance is governed by IRDA** which is an independent body. It has no direct connection to MoHFW, whereas the PMJAY comes under NHA.
- **Hospitals' standards are defined by NABH.** National Accreditation Board for Hospitals & Healthcare Providers [NABH] is part of the Quality Council of India [QCI] which is an independent body, but it has no powers to enforce. Less than 1% of Hospitals are accredited by NABH.
- Doctors' practices, medical education, and Primary care come under the Medical Council of India [MCI], now National Medical Commission [NMC]. This is an independent body. It has at best, a dotted line connection to the MoHFW.
- The Clinical Establishment Bill has been pending for discussion in parliament forever. National Identification Number for Healthcare Facilities of India [NIN HFI] and National Health Resource Repository [NHRR] have not been implemented across public and private sectors yet. Insurance Regulatory and Development Authority [IRDA] also has a ROHINI [Registry of Hospitals in Network of Insurance] database of its empaneled facilities. Furthermore, NHA has its database of impanelled Hospitals. Resolving this in a federated system can go a long way in addressing the big challenges discussed above by facilitating the easy exchange of Health Information through digital technologies.
- There is no regulatory control over - Emergency support, Technicians, GDAs, etc. The National Paramedical and Auxiliary Medical Staff Councils are still elusive. Radiology centres are controlled by Bhabha Atomic Research Centre [BARC], which is an independent body and comes under the Department of Atomic Energy. It has no connection to MoHFW. There is also no automatic reporting of import of radiological equipment to BARC which makes for a porous situation.
- Laboratory standards are defined by National Accreditation Board for testing and calibration laboratories [NABL]. NABL is part of the Quality Council of India [QCI] which is an independent body, but it has no powers to enforce. Less than 1% of Labs are accredited by NABL.

- Pharma sector is governed by the Central Drugs Standard Control Organisation [CDSCO] and Drugs Controller General of India [DCGI] which are independent bodies. It has a dotted line reporting to the Ministry of Chemicals and MoHFW.

Need for Interoperability.

Better care starts with interoperability - Interoperability in healthcare is crucial, with rapid technological advances. It is not only to embrace connected health and interoperability requirements - it is about improving patient outcomes. Interoperability is needed for healthcare organizations, to be able to ingest, manage, store, view, share and exchange patient data. It is the need of the hour to implement solutions that enable successful data exchange so that patients can stop ferrying their medical records from treatment to treatment.

The different vertical programs and directorates under MoHFW built their Healthcare-IT systems at different points in time using different technologies that were never designed for talking to each other. If a mother delivers a low-birth-weight baby and the mother has tuberculosis [TB] in the past, there is no way to trace if the low-birth-weight baby was due to the mother's TB, because each of the data points sits in different, siloed systems. The condition of the private sector is just as bad. None of the systems were ever designed for interoperability. Each application is developed for standalone use without much attention to semantic interoperability. Later when the thought of interoperability emerges – it becomes difficult to connect the systems and make them talk to each other because they were never designed for that purpose.

Even if technical and organizational interoperability is done the semantic interoperability may remain a challenge. For example – all applications must have the same facility master. When Application A sends the ANC data for Facility 123, the receiving Application B should understand ANC and uniquely identify Facility 123. Another example is, if a hospital application sends the insurance reimbursement bill to the insurance company/government, the recipient application should be able to understand and re-present the same meaning of bill information.

To address the challenge, Professor Dennis Streveler and Dr. Pankaj Gupta wrote the concept of eObjects for interoperability in Chapter 5 of the Health Systems for New India book published by Niti Aayog in Nov 2019. Later ACCESS Health Digital detailed out each of the eObjects, such that they comply with NDHB, EHR, and Metadata standards, guidelines provided by Medical Council of India, Clinic Establishment Act, and as per the latest telemedicine guidelines released by the Government of India. These eObjects are now published in opensource.

The eObjects produce machine-readable outputs such that the receiving facility can process the information without human intervention. They are designed to be inclusive, and suitable for adoption by a federated ecosystem, where different stakeholders could have deployed different technologies or be at varying levels of maturity in the development of their IT systems. The eObjects are therefore designed to be suitable for

initial deployment with or without FHIR standards, thus allowing for an evolutionary process so that no stakeholder needs to be left out.

The various eObjects are that have been published are as follows:

- **ePrescription** – Standard format for prescription to be machine-readable.
- **eReferral** – For sending a referral from primary to secondary to tertiary facilities.
- **eEncounter** – carries all the details of a clinical encounter in a machine-readable format.
- **eDiagnostics** – Lab and Radiology reports being machine-readable.
- **eDischarge** – Machine-readable discharge summary after an In-Patient episode.
- **eClaims and ePayment** - for the Electronic Health Claims Platform].

Document Registry: Digilocker

The use of Digilocker as a document registry with eObjects would help achieve even more efficacious interoperability in the ecosystem. It serves as a viable option to store the eObjects and share them with the payer or another provider. Professor Dennis Streveler along with ACCESS Health team propose a simple high-level solution architecture where Digilocker serves as a document registry to store eObjects as pointers to heavier attachments such as hospital records, lab results, and images which than can be stored in respective Health Repository Providers. The proposed approached was further discussed with Provider Working group during consultations.

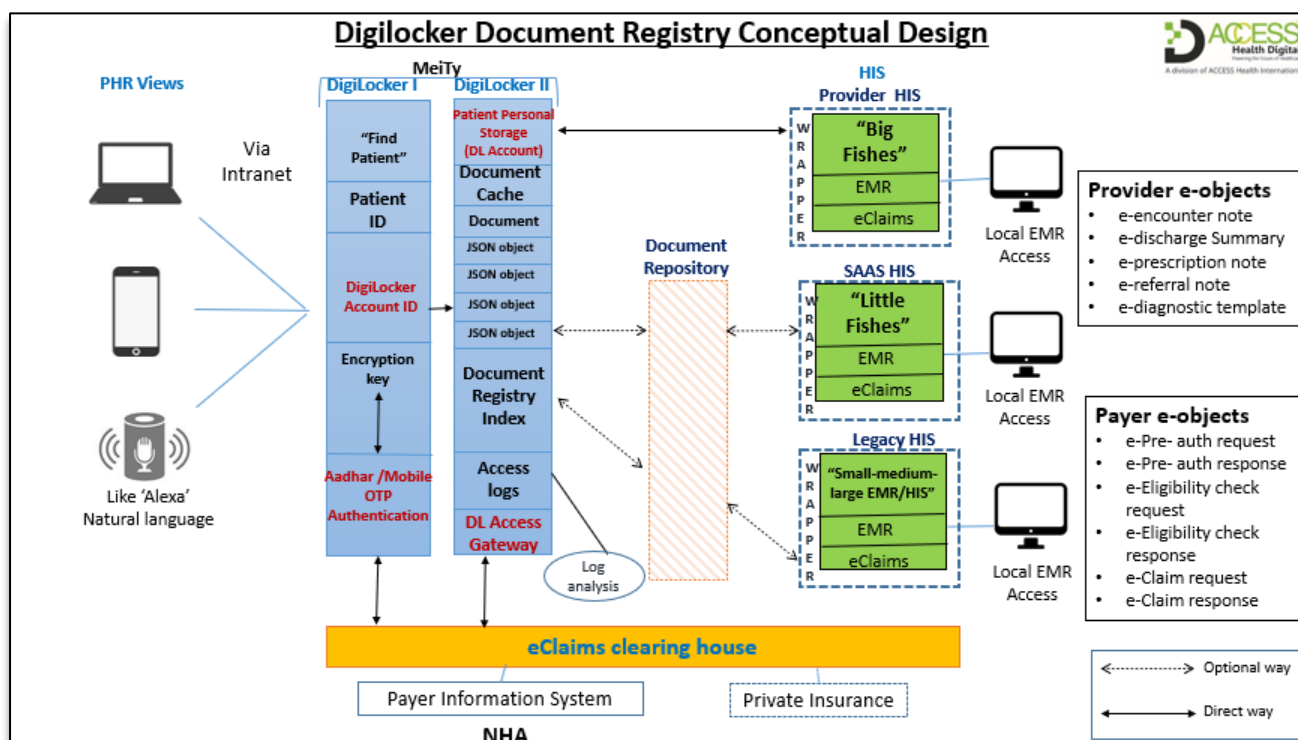


Figure 1: Digilocker Document Registry Conceptual Design

Benefits of Interoperability

Some of the benefits of interoperability are listed below:

- When doctors have access to complete patient data, they can make the best possible medical decisions and avoid duplicating tests and procedures.
- Interoperability can help reduce administrative costs and manual workflows while helping organizations improve efficiency and reduce medical errors. Reducing paperwork for staff and patients alike is a huge benefit that comes with interoperability.
- It allows building longitudinal patient records across EHRs and healthcare facilities.
- Interoperability enables connection to the larger group of providers and organizations – anytime, anywhere in the country.
- Effective health data exchange can also help to improve patient care coordination, reduce hospital readmissions, and ultimately save money.

PM Modi Ji's Iconic speeches at the launch of Ayushman Bharat and NDHM

"Over 50 crores of nation's brothers and sisters will be the recipients of health assurance worth up to Rs. 5 lakhs, in what is the world's largest scheme of its kind. Nowhere in the world will you find any government funding and running a scheme of this magnitude. The entire population of the European Union, that is the number of people in India who will benefit from the Ayushman Bharat scheme. I do not want you to fall ill, but if you do, there will be health and wellness centres next to you, and god forbid if you need hospitalization then PMJAY will be standing there to offer the coverage".

"National Digital Health Mission - a revolution in the health sector in India, starts today on 15th August 2020. Every Indian will get a health ID which will be a repository of all health-related information of a person.
#AatmaNirbharBharat"

Healthcare Industry – Provider Working Groups

Access Health Digital [AHD] formed three Healthcare Providers Working Groups to discuss and evolve a majority agreement towards the exchange of health information in standardized formats to foster a digital healthcare ecosystem in India. The objective of these working groups was to reach a majority-based agreement on sector-specific interoperability standards and present the results to the National Digital Health Mission [NDHM] for implementation and adoption of the National Digital Health Blueprint [NDHB] and thus creation of National Digital Health Ecosystem [NDHE].

The three Healthcare Provider Working group formed by Access Health Digital were:

- **Telemedicine Working Group**
- **Pharmacy Working Group**
- **Diagnostic Working Group (Laboratory and Radiology)**

Telemedicine Working Group

Telemedicine makes it possible to offer remote diagnosis and treatment to the patients, using electronic information and communications technologies and supports the provision of effective healthcare, when distance separates the participants or where capacity and resource constraints are the concern. Telemedicine holds the promise to improve access to care, improve patient satisfaction, and reduce costs to the health care system. Research shows that telemedicine can potentially reduce costs, improve health outcomes, and increase access to primary and specialty care. According to Global Telemedicine Market Analysis published by Market Data Forecast in 2019, the global telemedicine market is projected to rise to US\$ 80.61 billion by 2024 from US\$ 35.46 billion in 2019, growing at a CAGR of 17.85%. The telemedicine market in India is expected to reach US\$ 5.4 Bn by 2025 with a CAGR of 31%.

However, the full potential of these advancements cannot be reached without incorporating clinical and technical standards and appropriate guidelines. In the absence of standards and guidelines, interoperability and interconnection are not possible to achieve, and the great promise of telemedicine shall remain unfulfilled. The potential is immense, but big hurdles remain before it reaches its full potential.

Meaningful use of Telemedicine hinges on the ability for it to integrate with the ecosystem and exchange health data with it. Currently, it is being seen merely as a tool and not as a clinical setting by itself. Telemedicine is not merely about the video; it is the coupling of the clinical workflows and technology that is required to maximize the quality of the encounter.

ACCESS Health Digital formed the Telemedicine Working Group comprising of Healthcare Industry providers to arrive at a common consensus on Telemedicine standards, semantic interoperability, and guidelines for India, which are uniform across the ecosystem and can be integrated at the national level using NDHB based building blocks - Minimum Viable Product [MVP].

The Telemedicine working group gathered twice a week in an online discussion forum for 1hr/1:30hr, during Nov'2020 ~ Dec'2020 to deliberate upon the much-needed Telemedicine semantic interoperability standards and assessed the teleconsultation use cases catering to acute OPD, chronic disease management, prep for an in-patient procedure, follow-ups after procedure, rehab, step down care, home monitoring, and wellness. Following table shows the calendar followed for the Telemedicine Working group consultations.

TELEMEDICINE PROVIDER WORKING GROUP - Calendar				
Consultation Sessions				Governance Meeting
Kick-off Session	Wed, 28-Oct-20			Tue, 08-Dec-20
ePrescription Object eEncounter Object eReferral Object	Tue, 10-Nov-20	Fri, 13-Nov-20	Tue, 17-Nov-20	
	Fri, 20-Nov-20	Fri, 27-Nov-20	Tue, 01-Dec-20	
	Fri, 04-Dec-20	Fri, 11-Dec-20		
				# Sessions
				10

Towards the end of working group discussions, the summary was shared with the Governance Group that consisted of industry leaders for larger consensus on Telemedicine semantic interoperability standards.

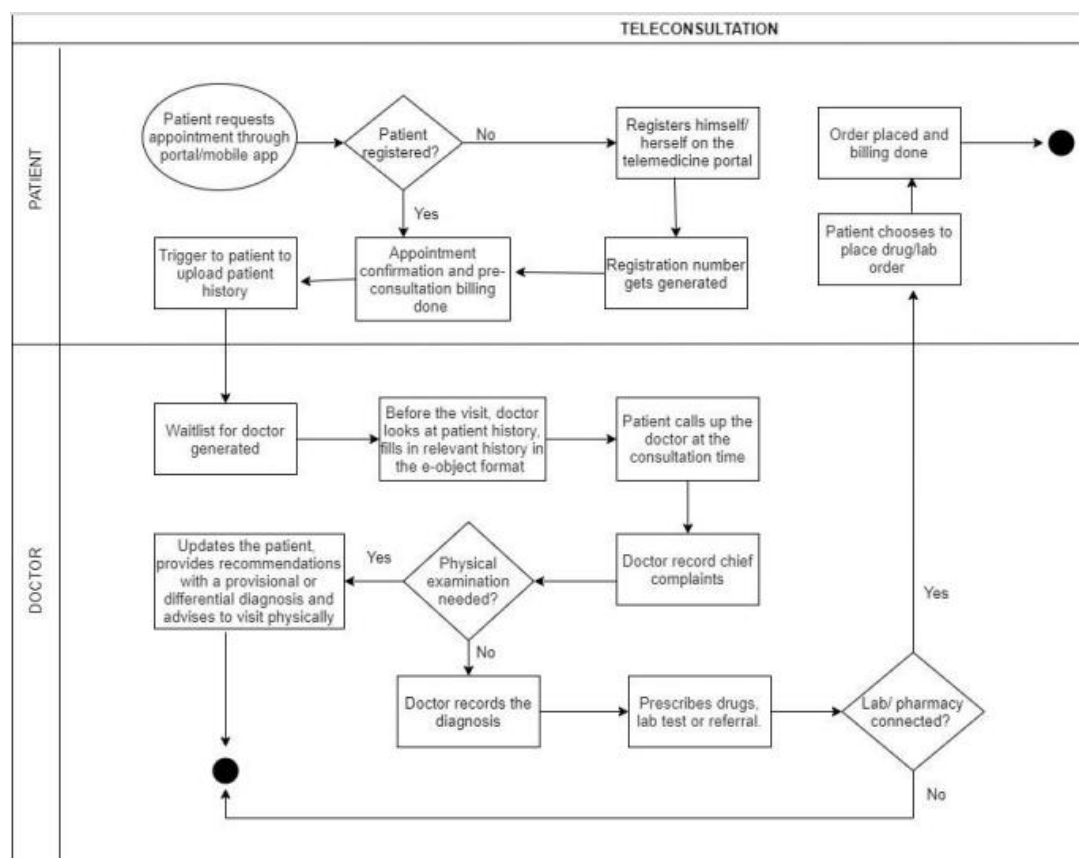


Figure 2: Teleconsultation Use Case

To support interoperability across disparate Telemedicine systems from various vendors, the group deliberated upon and agreed to the interoperability standards required to deliver effective telemedicine services across multiple systems and platforms in the ecosystem, keeping the following design principles in focus:

- Minimal Change to existing systems.
- Low Cost
- Minimal Change to existing workflows within an institution – market driven, phased approach to NDHB implementation.
- Benefits accrued through the aggregation of data – via ecosystem-based data science initiatives.
- Reduce medical costs when used as a component of a patient's longitudinal care.

Telemedicine Working Group - Recommendations:

The working group believes that telemedicine can be most efficient and beneficial between a patient and providers with an established, ongoing relationship and standard protocols. Extending the delivery of care model to have a virtual component does not require a redo of the health IT infrastructure, however, it does require Health IT vendors to be open and further permit a bi-directional interoperable information exchange with other IT solutions in the ecosystem including device integration.

The Telemedicine guidelines have put the responsibility on the Registered Medical Practitioner for maintaining sufficient medical records. The medical records can be physical or electronic format. For Interoperability, the use of the following eObjects and their Designs/ structures/formats was agreed upon by the majority of the members in the Telemedicine working group:

- **ePrescription Object**
- **eEncounter Object**
- **eReferral Object**

Details of these eObjects are provided in Appendix: [ePrescription eObject, eEncounter eObject, and eReferral eObject .](#)

The Telemedicine working group members also agreed on using the following standards, to create the much-needed interoperability among healthcare systems; minimizing the risks of new technology development; preventing single vendor lock-in; reduce costs by enabling market competition and eliminating the need for expensive and customized solutions and enabling ensuring widespread adoption.

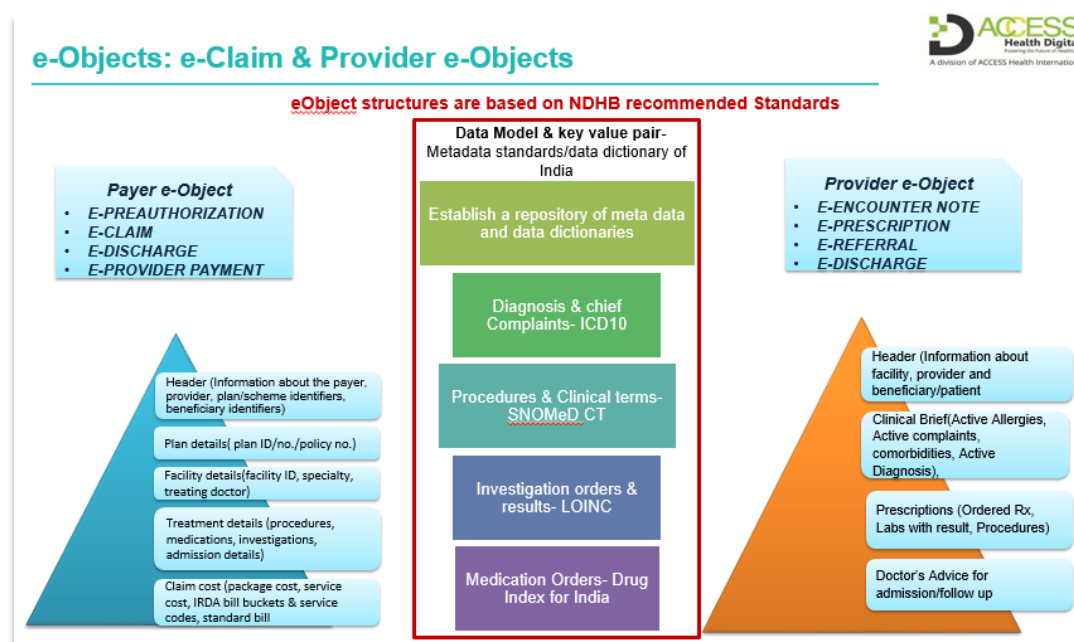


Figure 3: eObjects: eClaims & Provider eObjects

- **Metadata and data dictionary**, in principle, the working group agreed on the value of greater metadata use. Metadata and data dictionary tagging provides the potential to increase the usefulness and integrity of data for health information exchange by better characterizing the information being shared across the healthcare systems.
- **ICD10** (International Classification of Diseases, 10th Edition) for diagnostic purposes, classifying diseases and storing diagnostic information for clinical, quality, and epidemiological purposes, and for reimbursement of claims.
- **SNOMED-CT** (Systematized Nomenclature of Medicine -- Clinical Terms) for the procedure and clinical terminology, as it consists of sets of clinical phrases or terms, that can be grouped with relationships between terms.
- **LOINC** (Logical Observation Identifiers Names and Codes) for lab-tests, enabling the comparability and analysis of consolidated laboratory result data. It also facilitates increased access to laboratory test results across the continuum of care and reduces the need to repeat laboratory tests by providing a universal identifier for lab tests across care sites and systems.
- Most of the radiology diagnostic procedures are covered under **LOINC**. Please lookup **SNOMED-CT** for the radiology therapy procedures. In addition, it is recommended to use **DICOM** Standards for radiology images. DICOM coupled with HL7/FHIR eases interoperability across systems.
- Though the rules say that the prescription should be in generic drugs format however the physicians are comfortable having Brand preferences. The largest brand has not more than 8%-10% market share of a particular region in the country. In a Telemedicine consult, the prescribed brand may not be available in the patient's region. The same brand may have different active substances. Hence the

working group strongly recommends that there needs to be a **National Drug Index** for India, created, and maintained appropriately by the Government of India in partnership with the Industry.

The Working group believes that telemedicine can potentially be a beneficial and important part of the future of health care delivery in India, as telemedicine continues to become more of a “must-have” rather than just a “nice-to-have” platform.

As interoperability standards, specifications, and interfaces become more robust and more broadly implemented, they will support an even richer telehealth infrastructure and a seamless, scalable, improved clinical experience across the ecosystem. This will also be critical to the evolution of disease management templates over some time, which will make healthcare in India even more efficient.

The wider adoption of Telemedicine will require a business model to emerge around it. This will help it to graduate from being a tool, to a complete clinical setting for the delivery of care. This may also require a payment mechanism to be integral to telemedicine solutions.

Pharmacy Working Group

The pharmaceutical industry in India was valued at US\$ 33 billion in 2017 and generic drugs account for 20% of global exports in terms of volume, making the country the largest provider of generic medicines globally. Despite such vast growth potential, the pharmaceutical sector in India is highly fragmented and unregulated and has lots of fuzzy boundaries which impede maintainability and accountability. The challenge becomes more complex with the absence of an Integrated National Level Drug Index functioning as a single-source-of-truth.

The visibility that an authentic National Level Drug Index will offer, can transform the way the drugs and pharmaceutical sector is regulated and governed today. Simultaneously, the rules governing the drugs and manufacturers also need to be updated. Coherent resolution of all the existing drugs-related issues, calls for a single source of truth and a reference for all the drugs available across the nation.

To ensure quality, safety, efficacy, availability, and accessibility, a National Drug Index is of utmost priority. In the present scenario, the data available is highly fragmented and needs to be integrated into a centralized database at the national level. It is important that the provenance of data, be traceable too.

The second working group formed by ACCESS Health Digital was the Pharmacy working group comprising of representations from pharmaceutical manufacturing companies, distributors, organized pharmacies, and industry groups. The working group aims to serve as a communication platform between the pharmaceutical companies and government authorities and regulators.

The Pharmacy working group gathered twice a week in an online discussion forum for 1hr/1:30hr for 8-sessions, during Dec'2020 to deliberate upon interoperability standards and on how to form and maintain a structured National Drug Index, which in turn can address the jurisdictional loopholes that have resulted from the fuzzy boundaries discussed earlier in this report. The group also recommended that once National Drug Index is built, the drug regulators would need to maintain the drug index whenever they approve new drugs or restrict any drugs.

Following table shows the calendar followed for the Pharmacy working group consultations.

PHARMACY PROVIDER WORKING GROUP - Calendar					
Consultation Sessions				Governance Meeting	# Sessions
Kick-off Session	Tue, 15-Dec-20			Fri, 5-Feb-21	8
1) ePrescription Object	Tue, 22-Dec-20	Thu, 24-Dec-20	Tue, 29-Dec-2020		
2) Drug Index					
BCG – Discussion and feedback on the national drug registry with NHA	Tue, 20-Oct-20	Mon, 9-Nov-20	Tue, 19-Jan-21		

Pharmacy Working Group - Recommendations:

- The use of the following eObjects and their designs/ structures/formats was agreed upon by most of the members in the Pharmacy working group:
 - ePrescription Object

Details of these eObjects are provided in Appendix: [ePrescription eObject](#)

- Brand mapping and creation and maintenance of a National Drug Index.**

Standard drug index, a prerequisite: The requirement for a comprehensive, real-time information database on drugs and manufacturers persists within the country. The regulatory agency currently relies on the information gathered by state authorities, with a lack of abstract conceptual knowledge of what drugs are available in other states. Each state maintains their own databases which are not provided for integration at the national level for a centralized lookup.

Over the past few decades there have been numerous, very intensive, attempts at creating a National Drug Index for India, including the National Formulary of India, e-Aushadhi, Sugam portal, and numerous others. None of these have been sustainable or effective, for reasons that are myriad and too onerous to go into in detail. Suffice to say that the 'Pharmacy of the World' and the 'IT Capital of the world' does not have National Drug registry to support its healthcare system and provide SAFE and HIGH-QUALITY drugs to its citizens.

Given the above, detailed discussions were held in a Special Interest Group [SIG] on the National Drug Index for India including members from the pharmaceutical, pharmacy industry and industry bodies. The SIG strongly recommends that there shall be a National Drug Index for India, created and maintained appropriately by the Government of India in partnership with the Industry.

The National Drug Registry should maintain information on the following data elements, as a minimum data set-

- Drug Code
- Drug classification (Analgesic, antibiotics, etc.)
- Generic Name
- Active ingredients
- Brand Name/manufacturer name
- Manufacturer License details
- Drug Form (liquid, powder, etc.)
- Drug-Drug Interactions
- Dosage and Usage amount/information
- Route
- Alternate Drug
- Pricing
- Type of packaging

The drug index would also eliminate the instance of different generic drugs being sold in the market under the same brand name.

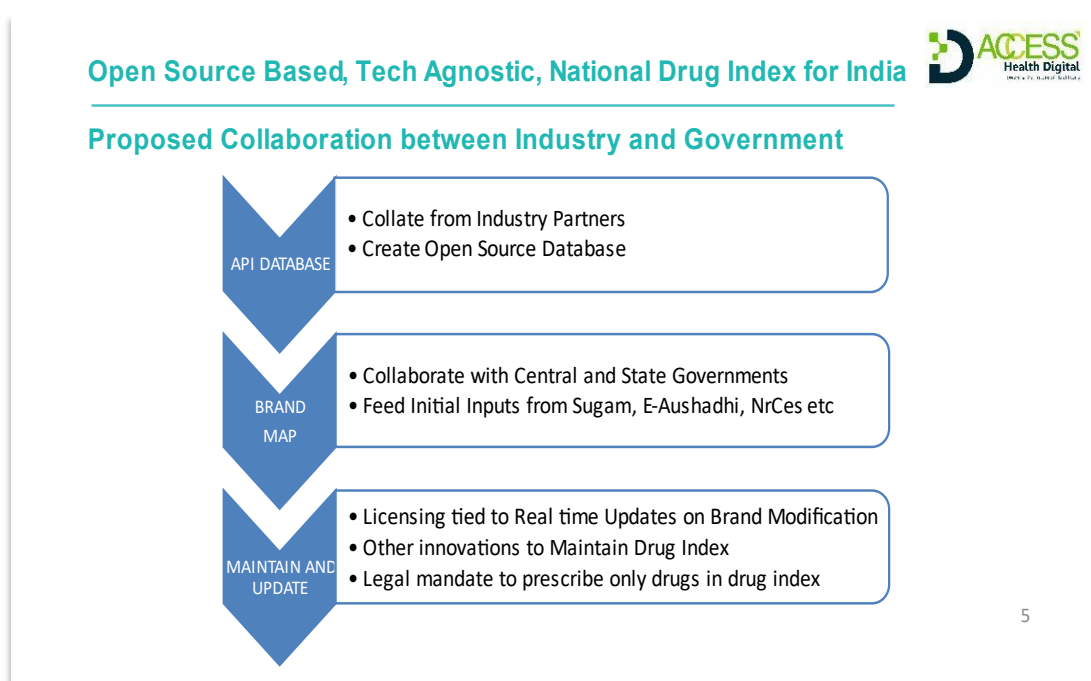


Figure 4: National Drug Index of India – Proposed Methodology

The following is a summary of the recommendations for the National Drug Index:

- The Drug Index must contain a comprehensive database of APIs and map the Brands to it. It should be based on the Opensource model; should be technology agnostic; should not be based on or hinging on any other Standard and should be created, owned, and maintained by the Government.
- Seed the creation of the proposed National Drug Index through an initial comprehensive compilation of basic API items from global public and private entities. Organizations from within the industry stand ready to collate and publish this as an open-source database created for all and owned by the government.
- A process for initial mapping of brands to the API's including specifying the structure of such a mapping.
- A process for ensuring that ANY changes - including creation, stopping, banning, changes to active ingredients - of brands by a manufacturer anywhere in the country is logged through one or more portals and triggers an update (automatic or manual) of the Drug Index.
- A policy mandating that without the presence of a brand in the National Drug Index, it would be against the law to sell or prescribe it. Mandate use of Bar Codes from manufacturing to point of care.

References can be drawn from across the world, for instance, in the United States, a National Drug Code Directory is maintained as an open-source directory with all the available drugs approved and provided by FDA. USA also possesses First Data Bank (FDB) providing an integrated drug database and is the leading provider of clinical and descriptive drug knowledge, which is further integrated into healthcare information systems globally. In the United Kingdom, the NHS (National Health Services) provides a list of medicines and their associated information in a form of a drug index at the national level, similarly, among ASEAN countries Philippines has a drug database provided by the FDA of the Philippines.

To safeguard the health of the people of India there is an absolute need for a standardized National Drug Index which in turn can cover up the jurisdictional loopholes that have resulted from the fuzzy boundaries. Once built, the Government must own it, the drug regulators appointed by the Government would need to maintain the drug index whenever they approve new drugs or restrict any drugs.

Diagnostics Working Group

The field of Lab-diagnostics faces many challenges including the adoption of new technology, new ways of performing tests, and sharing the data. It is important to establish the collaboration between lab professionals, manufacturers, and physicians with regards to the digital health platforms and agreed upon input-output standardization that involves a standard list for test panels. If we are to unlock the true potential of the digital lab workflow, it is essential to promote the coordinated use of input-output standards to achieve interoperability between various systems and the effective use of electronic health records (EHRs) – the proposition is to have “Metadata-Enhanced Workflows.”

ACCESS Health Digital formed the third working group - Diagnostics Working Group comprising of Industry professionals/experts from hospitals, laboratories, diagnostic organizations, and technology companies. Currently, there is no single set of standards that can help seamlessly exchange health data. Each local system and organization have their own set of data standards or no standards. This is the reason, why data is isolated and not interoperable at the diagnostic level.

The Diagnostic working group deliberated on the eDiagnostic template eObject and Lab/Diagnostic Panels. The sole purpose for the formation of the Diagnostic Working Group was to resolve the issue of interoperability by identifying the standards in clinical laboratory and radiology data exchange.

The Diagnostic working group gathered twice a week for 9 - sessions in an online discussion forum, during Jan'2021~Feb'2021 to deliberate upon interoperability standards and certain initial conclusions and recommendations were drawn up. Based on these inputs and interactions during the discussions, the working group constituted a focused Special Interest Group (SIG) that further deliberated about the final set of recommendations, to achieve a single set of standards and codes for easier and semantic interoperability of individual health records between various systems.

Following table shows the calendar followed for the Pharmacy working group consultations.

DIAGNOSTIC PROVIDER WORKING GROUP - Calendar				
Consultation Sessions				Governance Meeting
				# Sessions
Kick-off Session	Fri, 08-Jan-21			Fri, 05-Feb-21
1) eDiagnostic Object 2) Standard List of Test Panels 3) Standard List of Radiology Therapy Procedures	Tue, 12-Jan-21	Fri, 15-Jan-21		
SIG - Discussion on Lab and Radiology Panels	Tue, 19-Jan-21	Fri, 22-Jan-21	Mon, 25-Jan-21	
	Fri, 29-Jan-21	Tue, 02-Feb-21		

The Diagnostic working group recommended the use of LOINC for test orders and results. LOINC is a common language (set of identifiers, names, and codes) for identifying diagnostic measurements, observations, and documents and globally used by labs, healthcare providers, government agencies, to name a few. It helps to map the local tests (individual health data) to a national/global standard that every other local system can understand.

The SIG reached a consensus amongst the group on the directory structure for laboratory & radiology of the e-Diagnostic Object.

Diagnostic Working group – Recommendations:

- The Broad Diagnostic Working Group debated extensively on the following:
 - a) eDiagnostic Object for syntactic and semantic interoperability of Laboratory and Radiology Results across Providers, Labs, and Patients. The core discussions were around the minimum lab and radiology results-related data that would need to be made available to the clinician and patient to enable high quality, low-cost care to be delivered optimally at the point of care. Topics that were debated included the importance of some data on test methodology to be included in the eObject.
 - b) Laboratory and Radiology Panels – It was agreed by all participants that there is a crying need for basic standardization across the country on at least a bare minimum set of ‘standards panels’ for Lab and Radiology related to basic events/conditions / clinical diagnosis. This would be the absolute minimum that would be necessary to kick off an ecosystem-based set of initiatives around actual on-the ground implementation and automated monitoring of clinical guidelines.
 - c) Special Interest Groups (SIG) were formed across private labs, radiologists, aggregators, hospitals, and lab information system providers to generate these recommendations.
 - d) LOINC was agreed upon as the base data standard to be followed for semantic interoperability in this area.

- The use of the following eObject and their designs/ structures/formats were agreed upon by the members of the Diagnostic working group:

- **eDiagnostic Object**

Details of the eObject are provided in Appendix: [eDiagnostic eObject: Laboratory Template, and eDiagnostic eObject : Radiology Template](#)

- The lab SIG agreed that there were wide variations in lab panels between different establishments, and thus using LOINC needs to be nuanced to the Indian context. To do so, commonly ordered lab panels were listed out, as gathered from the Government sector e.g., CGHS and various private sector laboratories, lab aggregators, and lab information systems. The SIG applied the 80-20 principle to extract the most ordered Lab Tests and Lab Panels.
- Based on these extracted panels, a minimum set of tests that were both clinically relevant and optimally available, in the context of India’s infrastructure, were defined as ‘standard’ panels for the country. Each such panel was given a LOINC or LOINC like code and the tests within each panel were standard LOINC coded tests. This would allow easy addition of the proposed new panels to the LOINC international standard if so desired in the future.

- These LOINC ‘panel’ codes would be put into the same code directory as the current set of LOINC codes defined in MeitY’s MDDS for Healthcare Code Directory.
- Any Laboratory that would perform additional tests in their local system under their local panel with similar clinical / lab goals – would be separately communicated to needed external systems as individual LOINC test codes. E.g., The ‘standard’ recommended ‘CBC by automated count panel’ would not have ESR in the set of tests included. If a Laboratory did do ESR as part of its service offerings under this panel, from a business perspective – then ESR would have to be communicated separately from the panel, during standard interoperability-based communications between 2 parties. Please see the data exchange diagram at the end of this section where a similar example is provided for communication of standard and non-standard ‘renal function test’ panels.
- The Radiology SIG only specified a similar structure for the Radiology Study Description Database, without going into the specific minimum Radiology Studies to be included. This detail will be worked on in the future and presented in the same manner as the standard lab panels.
- The SIG recommends the following database/tables/structures / formats and alignment to MeitY’s MDDS for healthcare open standards.

Table 1: MDDS Entity – Code Directory Relationships

Entity	Entity No	Attribute	Attribute Code	Related MDDS Code Directory
Lab	21	Lab Order Code	05.021.0022	05.024
Radiology	22	Radiology Procedure Code	05.022.0008	05.043
		Lab Panel Code	05.021.0022	Panel Code Directory
		Radiology Study Code	05.022.0008 Or New Attribute in Entity 22	Study Description Directory

Description of the above table:

- *Entity and Entity Number:* Top level grouping of Metadata in MeitY MDDS for health
- *Attribute and Attribute Code:* One attribute of an entity and it is code in MDDS. Please note that the attribute code for a test and a panel are the same i.e. 05.021.0022. In essence, it means that a Lab Panel is just another Lab Order with respect to the MDDS standard.
- *For the Radiology Study Attribute* – this could either be the same code as the Radiology Procedure code (i.e., 05.022.0008) or a completely new attribute of the radiology entity in MDDS.
- *Related MDDS Code Directory:* Associated (MDDS code directory) value set for a field / field code in MDDS Code Directories.

Table 2: Lab Panel Code Directory Structure.

Panel Code Directory Structure						
MDDS Additions in Related MDDS Code Directory		Panel Standards Database				
Panel Code (in CD05.024)	Panel Description (in CD05.024)	Disease/ Event/ Guideline Code	Panel Member LOINC Code	Panel Member Description	Version	Date/Time Latest Change
Add in CD05.024	Add in CD05.024	ICD 10. CD05.022	CD.05.024	CD.05.024		

Description of the above table:

- *MDDS Additions in Related MDDS Code Directory*: Additional 'codes' (values) to be added in the existing MDDS code directory indicated.
 - *Panel Standards Database*: Additional Code Directory to be maintained separately and added to MDDS when possible.
 - *Panel Code and Description*: LOINC or LOINC like code and 'standard' description FOR A PANEL to be added into CD05.024 MDDS code directory.
 - *Disease / Event / Guideline Code*: Any MeitY Open Standard code for a disease/event / guideline for which the panel is applicable. To be maintained in a separate Disease / Event / Guideline Code Directory in MDDS such as ICD 10 etc.
 - *Panel Member LOINC code and Description*: Set of 'tests' – codes and descriptions - from LOINC standard test codes that form part of a single panel.
 - *Version and Date / Time of change*: For versioning and audit trail purposes.
- CD05.024: LOINC or LOINC format code directory in MDDS.
- CD05.022: ICD 10 code directory in MDDS.

Table 3: Examples of LOINC for the Lab Panels: Lipid Profile

Panel LOINC	Panel Description	Panel Member LOINC Code	Panel Member Description
24331-1	Lipid 1996 panel - Serum or Plasma	70554	Cholesterol [Mass/volume] in Serum or Plasma
		2571-8	Triglyceride [Mass/volume] in Serum or Plasma
		2085-9	Cholesterol in HDL [Mass/volume] in Serum or Plasma
		13457-7	Cholesterol in LDL [Mass/volume] in Serum or Plasma by calculation
		13458-5	Cholesterol in VLDL [Mass/volume] in Serum or Plasma by calculation
		11054-4	Cholesterol in LDL/Cholesterol in HDL [Mass Ratio] in Serum or Plasma
		2896375	Cholesterol. Total/Cholesterol in HDL [Mass Ratio] in Serum or Plasma

Table 4: Basic Metabolic Panel

In the below example of Basic Metabolic Panel, there is a parameter that is also a Panel in itself.

Panel LOINC	Panel Description	Panel Member LOINC Code	Panel Member Description	Panel Member LOINC Code	Panel Member Description
51990-0	Basic metabolic panel - Blood				
		2339-0	Glucose [Mass/volume] in Blood		
		6299-2	Urea nitrogen [Mass/volume] in Blood		
		38483-4	Creatinine [Mass/volume] in Blood		
		49765-1	Calcium [Mass/volume] in Blood		
		55231-5	Electrolytes panel - Blood	2947-0	Sodium [Moles/volume] in Blood
				6298-4	Potassium [Moles/volume] in Blood
				2069-3	Chloride [Moles/volume] in Blood

Table 5: Radiology Code Directory Structure.

Radiology Study Description Directory Structure						
MDDS Enhancements		Radiology Study Description Database				
Radiology Procedure Code	Radiology Procedure Description	Disease / Event / Guideline Code	Study Code	Study Description	Version	Data/Time Latest Change
CD05.043	CD05.043	ICD 10. CD05.022	05.043 or New Code Directory	05.043 or New Code Directory		
Disease Code Directory						
Disease / Event Code	Disease Name					
ICD 10. CD05.022	ICD 10. CD05.022					

Description of the above table:

- *MDDS Enhancements*: Additional 'codes' (values) to be added in the existing MDDS code directory indicated.
- *Radiology Study Description Database*: Additional code directory to be maintained separately and added to MDDS when possible.
- *Radiology Procedure Code and Description*: Procedure code and 'standard' description FOR A STUDY to be added into CD05.043 MDDS code directory or A new code directory to be defined in MDDS for this purpose.
- *Disease / Event / Guideline Code*: Any MeitY open standard code for a disease/event / guideline for which the panel is applicable. To be maintained in a separate Disease / Event / Guideline Code Directory in MDDS such as ICD 10 etc.

- **Radiology Study Code and Description:** Study code and 'standard' description FOR A STUDY to be added into CD05.043 MDDS code directory or A new code directory to be defined in MDDS for this purpose.
- **Version and Date / Time of change:** For versioning and audit trail purposes.

CD05.043: Radiology Procedure code directory in MDDS.

CD05.022: ICD 10 code directory in MDDS.

Reference to National Committee on Metadata and Standards for Health Domain – Part III-Code Directories by Ministry of Health and Family Welfare.

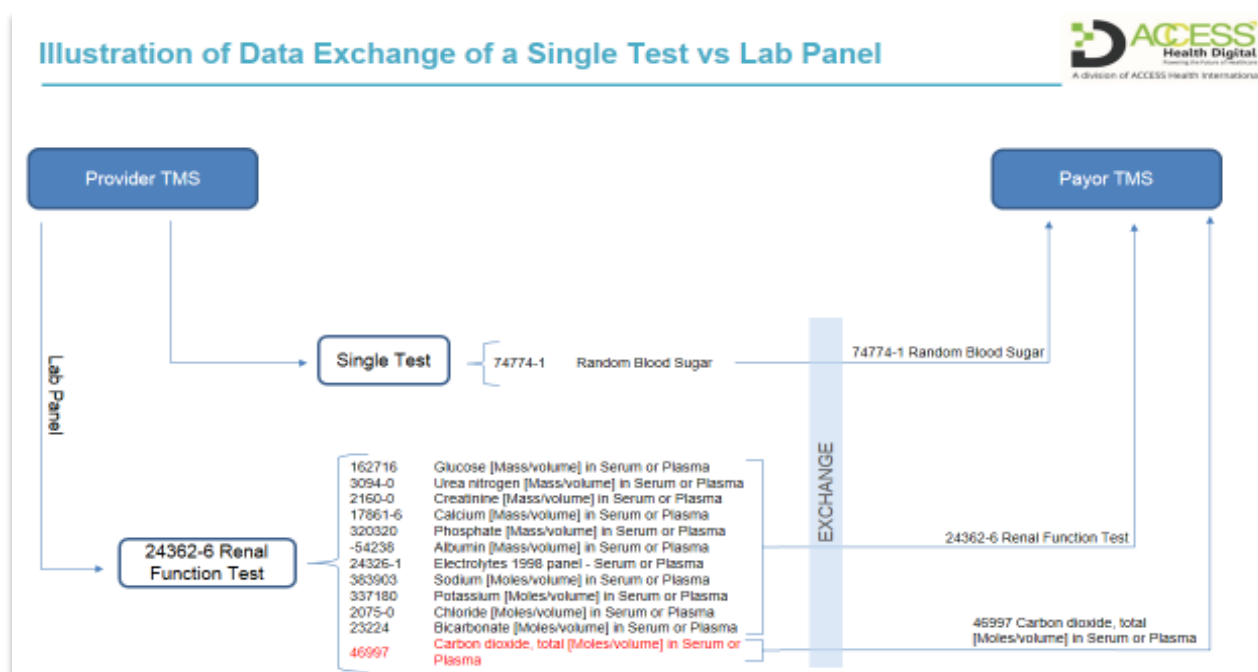


Figure 5: Lab Panel Data Exchange Example

NATHEALTH –Digital Health Breakout Sessions

NATHEALTH convened Digital Health Breakout sessions on 21-Dec-20 and on 24-Dec-20 respectively with ACCESS Health Digital as a Moderator, to discuss digital healthcare standards, adoption challenges, and interoperability with larger provider industry group. Participants were present from a range of healthcare areas like public health, NASSCOM, leading hospitals, major diagnostic labs, digital health providers, home healthcare players, financial organizations, healthcare consultancies, medical technology providers, and more. Breakout sessions were focused on:

- **Removing resistance to change & resolving intra-operability issues.**
- **Aligning track with NDHM & Formulating standards**

These breakout sessions were valuable in validating the assumptions and findings of the respective working groups and set the stage for furthering the cause of National Digital Health Ecosystem.

The group jointly went through the interoperability standards, the use of the following eObjects and their Designs/ structures/ formats, and interfaces; they agreed on the need for a seamless, scalable, and improved clinical experience across the ecosystem.

- **ePrescription Object**
- **eEncounter Object**
- **eReferral Object**
- **eDiagnostic Object**

Details of these eObjects are provided in Appendix: , [ePrescription eObject, eEncounter eObject, eReferral eObject, and eDiagnostic eObject: Laboratory Template](#)

eDischarge Summary Discussion Group

India is rapidly moving from a pay-out-of-pocket model to a Universal Healthcare Coverage model. It is in the interest of the provider sector to comply with the change, as the Insurance will soon start asking for eClaims in a standard format supported by eDischarge eObjects for a faster turnaround on in-patient claims. Public Sector insurance, Private Sector insurance, and Regulators including NHA and IRDA have agreed to adopt the eClaims and eDischarge eObjects for the new proposed Electronic Health Claims Platform. Please refer to the IRDA-NHA joint working group report.

The old HIS and EMR systems or even the new breed Digital Health Apps lack credible Global Digital Health Standards. Hence, they exist in silos and do not interact with each other or the larger Healthcare ecosystem. Interoperability among e-Governance HMIS applications for the health sector requires the exchange of in-patient information across HMIS applications. There is a need for standardised and commonly accepted data definitions for the various data elements used for e-Governance in HMIS systems across the Health System. The standardized in-patient data elements are bundled into the eDischarge eObject. Hence eDischarge eObject becomes a prerequisite for semantic interoperability across the Health System. eDischarge eObject was discussed with Provider Working group members on 16-March-2020 to update them on eDischarge interoperability standards, already adopted by the Payer Working Group.

When the pandemic hit us in 2020, we all were caught completely off guard. Semantically coherent machine-readable in-patient data could not be culled-out from existing HMIS/HIS/EMR systems for any meaningful

epidemiological analysis. We are still struggling with the epidemiology of the Covid-19 due to a lack of semantically coherent machine-readable in-patient data. The pandemic has not gone away yet and this isn't going to be the last pandemic we are facing. In addition, we still do not know the size, scale, and depth of the Dengue, Chikungunya and H1N1, Flu epidemics that strike us every year. Hence, we are always left gasping for breath when the seasonal spike starts. India is already the diabetes capital of the world with 7-Crore cases and still counting. We still do not have standard protocols based on the Digital Disease Management Platform.

Hence the HMIS/HIS/EMR market in India requires Digital Health standards-based FHIR/JSON/XML eDischarge e-Objects that can help the existing systems to communicate in-Patient information with the external world in a standardized format.

For Quick Reference, the details of the eDischarge eObjects are provided in Appendix [eDischarge eObject](#), [ePrescription eObject](#)

Conclusion

The idea of 'One Nation One Health' strives to offer Proactive or wellness-centric, Accessible, Equitable, Safe, and Portable healthcare to all Indians, and moving away from pay-out-of-pocket models. For Governance, payer, provider, and beneficiary to play their parts to their fullest part in a multi-polar interdependent ecosystem, a seamless, interoperable healthcare information ecosystem needs to emerge.

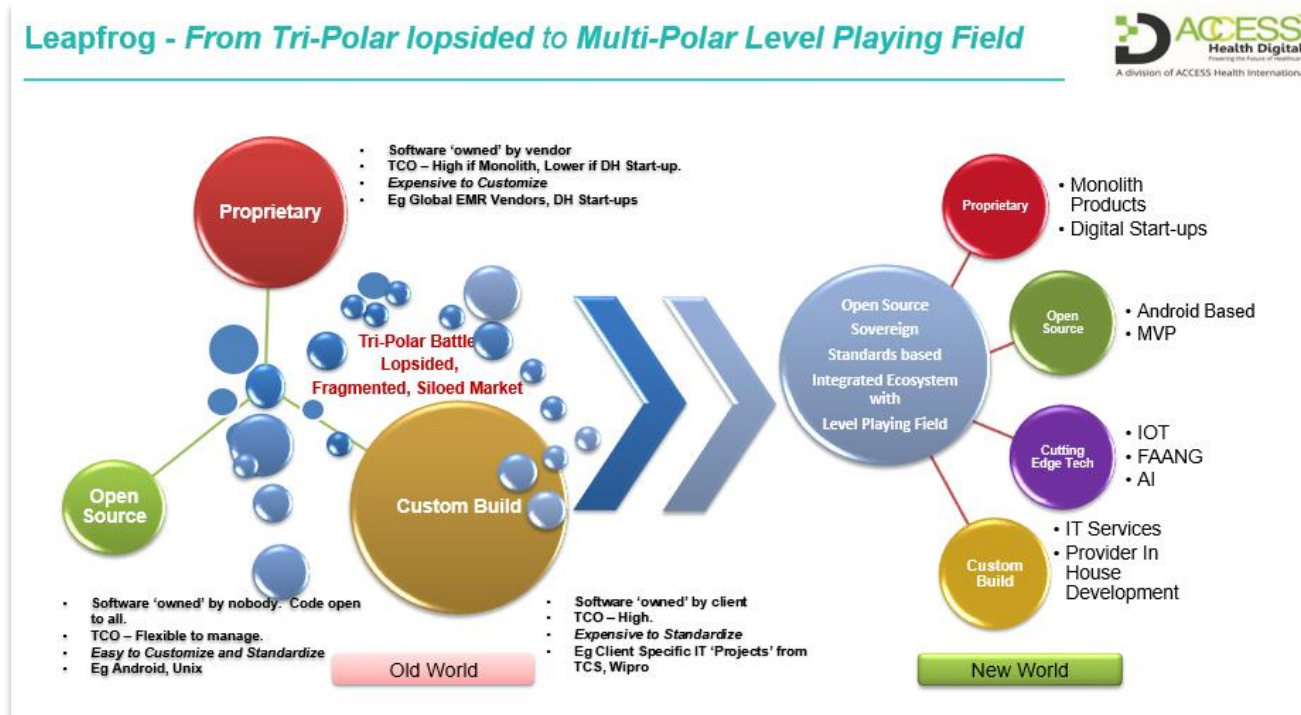


Figure 6: Level Playing Field – Bazar Model

Digital Innovation based on appropriate standards, building blocks, and a minimalistic approach is the way forward for India to leapfrog from its legacy of inadequate, disjointed, and disparate information systems to create an open, inclusive, interoperable ecosystem to meet this pressing national need.

Use India's opensource Metadata standards notified by MeITY and to have a level playing field that is tech agnostic, opensource, free, India Specific Standards.

The Provider Side Working Groups convened by ACCESS Health Digital aimed to achieve consensus on foundational elements of such an eco-system and elicit what the industry players feel would contribute to a system that can meet these lofty policy goals that the nation has set for itself in line with the universally accepted sustainable development goals framework.

Industry organizations, businesses, and thought leaders rose to this clarion call from ACCESS Health Digital in an unprecedented fashion, taking out time from their demanding schedules which were further stretched by the demands that Covid19 placed on them as organizations and individuals. Their sustained participation from Nov'2020 thru February 2021 validated the shared belief that the healthcare industry can place service above self and shares a deep concern for the Indian citizen.

To each one of these fine individuals, we owe deep gratitude and thanks. It may not be possible to thank all of them individually, however, some notable names are mentioned in the acknowledgment section and to the list of participants appended to this report.

This report spells out the challenges, detractors, and approaches to a National Digital Health Ecosystem [NDHE] that are critical to Universal Healthcare becoming a reality in India. This has historically taken most nations upwards of two decades. We believe it is possible to do this much faster in India with a consensual approach and following the recommendations that have emerged from this detailed and painstaking and inclusive undertaking to define the 'HOW' aspect of the NDHE. Individual sections on each of the three working groups have brought out recommendations that are paramount to success, in some detail. There are other relevant lessons from global experiences in digital transformations and the innovation models that have been used in areas of Public Good.

The Healthcare Systems of the mature economies have been in serious question with healthcare becoming increasingly unaffordable for both individuals and the public resources that it has ended up consuming to provide healthcare. This has much to do with the business model adopted. As a metaphor, this model can be compared to a "Cathedral" where entry is based on exclusion and discourages competitiveness and innovation. The 'Cathedral' model results from proprietary components finding their way into public systems. These proprietary components work as a gateway and are exclusionist.

The erosion of primary care in these systems is a notable result of the Cathedral model.

The Cathedral model is rigid and acts as a strong disincentive to competition and innovation. Its sustainability, maintainability is poor, and it creates dependencies and breeds obsolescence, which are detrimental to the public good. This is of course also more open to abuse.

Ecosystems are elastic, dynamic, and open by nature. They allow for growth and are decidedly inclusive. They can be compared metaphorically to an Indian 'Bazaar', given the diverse range of stakeholders performing various functions. Yet the Bazaars can fold-in these diverse stakeholders based on an agreed minimum set of rules to be followed. The inter-dependencies of these stakeholders are easily realized in such an environment.

The Digital Open-source innovation models are similar to the 'Bazar' where there a minimum set of common standards and rules agreed upon. This fosters interdependence, and yet individual needs are easily met. It also ensures that the majority of stakeholders can work together based on trust, transparency, and a level playing field assured by commonly accepted standards and rules. These are democratic environments that they can enter, or leave based on their individual choices without seeking any permissions. They mirror sovereign individual rights assured in a constitution, unlike rigid dictatorial regimes. Freedom of course comes with certain voluntary obligations. The widely sourced innovation keeps the system evolving and is an antidote to obsolescence and dependencies that are fatal.

India would do well to be mindful of these dynamics that are endemic to these models of ecosystems. The eObjects (ePrescription, eEncounter, eReferral, e-Discharge and eDiagnostics) represent provider industry consensus on their design and are the key enablers to Health Information Exchange across stakeholders. They can bridge over the diversity, fuzzy boundaries, and challenges that have been discussed in this report.

A National Drug Index is critical to accomplish National Digital Health Ecosystem and to ensure accessible, equitable, safe healthcare to a vast nation. Besides, we must also strive to create capacity both in terms of knowledge and infrastructure to ensure sustainability and maintainability of a National Digital Health Ecosystem [NDHE] that must serve the country in perpetuity. This, we must be mindful of, on the road ahead!

**Market Diversity can only be addressed through
Inclusivity.**

Respect the culture and adoption will happen.

Invitees List

Telemedicine Working Group

- Dr. Krishna Reddy, Country Director, Access Health International
- Dr. Sneha Kannan, BDS, Dentistry, iCliniq
- Dr. Ajay Alexander, Head - Medical, Practo Technologies Pvt. Ltd.
- Dr. Pankaj Gupta, Head-Digital Health, ACCESS Health Digital
- Dr. R. Balaji. Healthcare Expert, ACCESS Health Digital
- Dr. Rajesh Gupta, VP - Clinical Transformation at MyHealthcare Technologies
- Dr. Surajit Nundy, CEO Raxa
- Mr. Amarjeet Cheema, Joint Director, C-DAC Noida
- Mr. Ajay S Sharma, Co-Founder and Managing Director, Srishti Software Applications Pvt. Ltd.
- Mr. Ajit Narayanan, CTO, Mfine
- Mr. Dilip Jose, CEO, Manipal Group
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- Mr. Sudhanshu Gupta, Star Imaging & Path Lab
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- Mr. Umesh Rai, COO, Metropolis Healthcare Ltd
- Mr. Vijender Singh / Ameera Shah, CEO / MD, Metropolis Healthcare Ltd
- Mr. Vishal Nagpal, Co-Founder, ITDOSE Inforsystems Pvt Ltd
- Ms. Rituparna Sen, Lead - Patient Care Service, BM Birla Heart Research Center

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- Mr. VG Somani, Drug Controller General of India
- Mr. Vijendra Singh, CEO, Metropolis Healthcare Ltd.
- Mr. Yateesh Wahal, Director, Nayati Healthcare Pvt. Ltd.
- Mr.KG Ananthakrishnan, Director General, Organization of Pharmaceutical Producers of India.
- Prof. Dr. N.K. Ganguly, Co-Chairman, Indian Pharmacopoeia Commission

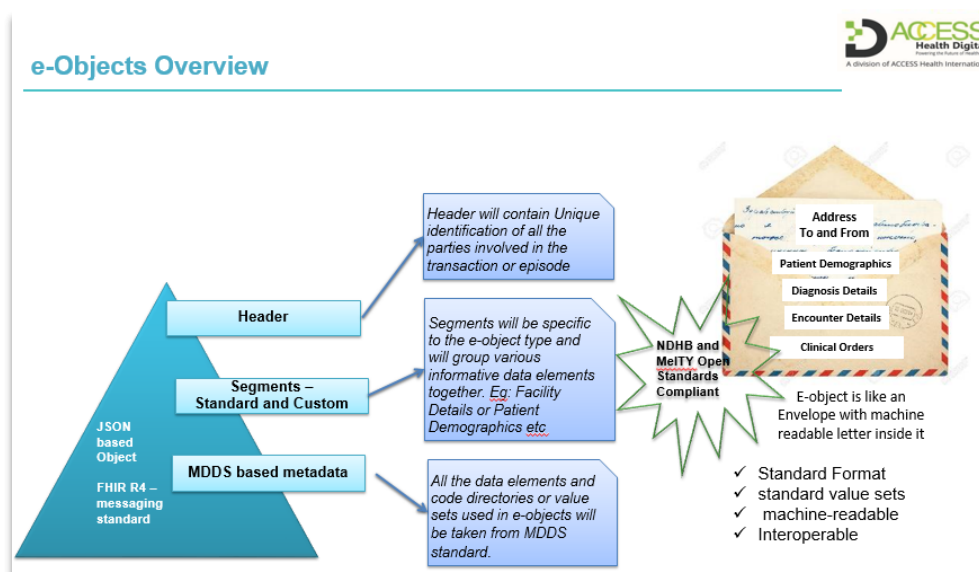
Annexures

eObjects Introduction

Electronic objects (e-Objects) are a system goal-specific container for various types of health documents. E-objects are digitalized versions of the document along with associated values and value sets with the format, size, and structure of each field.

The e-Objects are like an “envelope of information” with an address of the receiver and sender to route the object to the right stakeholder (Provider/payer/beneficiary) and have a detailed body to carry relevant protected information about an individual. Each electronic object that will be generated or gets created during a cycle will carry a unique identification number. All the e-Objects that are generated for one cycle should be linked to a case ID that can bundle the relevant e-objects together and streamline the workflow.

The structure of the e-Objects as recommended by the group will consist of two parts:

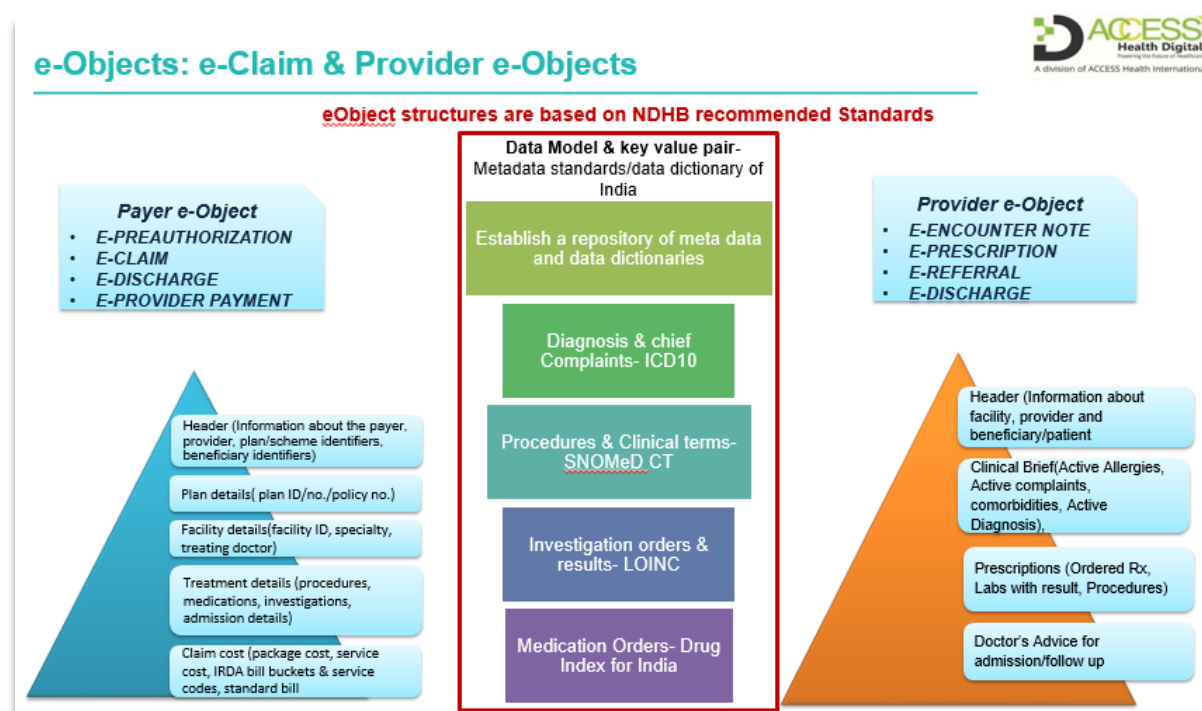


Header- Acts as the address and will be encrypted by the requester on the healthcare platform using a public key. This part consists of unique identification numbers of all the entities that are involved in the transaction, linking them to the respective registries (beneficiary, payer, provider). These registries serve as a single source of truth for any required information by a payer or provider to request or to facilitate the reliability of data being exchanged.

Detail - The detail section will be encrypted and will only be read by an authorized stakeholder (Payer or Provider) using its authenticated private key. The detail section contains all the minimum data sets, health conditions, and service delivery information for a beneficiary, which is required by the provider to view the individual history.

With these emerging opportunities, the time has come for the Healthcare sector to leapfrog and be ready to transform through digital innovations into an Interoperable National Digital Health Ecosystem with standardized processes, definitions, and data sets. The industry Provider group came together, deliberated extensively and agreed to the following e-objects specifications for Telemedicine, Pharmacy, Diagnostic segments:

- e-Encounter Note Object
- e-Referral Object
- e-Prescription Object
- e-Diagnostic Object
- e-Discharge Object
- e-Claim Object (Financial Lever)



The design specifications of these e-Objects as well as some implementations of these, in a few different technology platforms, are available as open-source digital assets for India's public good by ACCESS Health Digital at the following location:

- <http://www.openbodhik.in/2020/04/provider-eobjects-published.html>
- <http://www.openbodhik.in/2020/08/eclaims-payer-eobjects.html>

ePrescription eObject

e-Prescription eObject will carry all the information of the prescription order sets given to the patient. It carries information pertaining to the medication part of the consultation process from the medication's orders, route of administration, refills and medication dispensations. All these are considered while creating the e-prescription object.

Data Elements Labels	FHIR Label	Field Type	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Header (To be encrypted and store)							
Unique Health Identification Number (UHID)	Patient.identifier	Optional	Integer	12		G01.01	Format
Unique identification (UID)	Unavailable	0 to 1	Integer	12		G01.01	To be used for ADHHAR (UIDAI) if authorized, This is used for person identification.
Alternate Unique Identification Number (UID) Type	Identifier.type	0 to many	Integer	12	CD05.007	05.002.0001	Till the time the UHID is unavailable, patient alternate id can be used in place of a unique identifier. For a hospital setting it can be the patient's AADHAR number. and for those enrolled in vertical programs, the IDs allotted to each patient under that program can be utilized for example, NIKSHAY ID. This can be extended to accommodate identifiers from various vertical programs for the same patient until we have UHID and a National Patient Master Index
Alternate Unique Identification Number (UID)	Person.identifier	0 to many	Varchar	254	Max.Size=18 10-PAN Card08 - 08-PassportNo. 18 -VoterID 18 -Any other Identifier	05.002.0002	
Unique Facility Identification Number (Generating)	Organization.identifier	0 to many	Mandatory	Integer	10	05.008.0001	If the prescription was generated by a healthcare facility, its ID from facility registry is to be captured

Data Elements Labels	FHIR Label	Field Type	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Unique Facility Identification Number (Receiving pharmacy)	Organization.identifier	0 to many	Mandatory	Integer	10	05.008.0001	
Order Verifying Care Provider ID	Composition.attester	Optional	Varchar	18		05.025.0006	Treating or prescribing doctor's State or MCI no. until Doctor's Registry goes live
Clinical Document Type Code	Composition.category	Optional	Integer	2	CD05.046	05.019.0006	
Prescription ID	MedicationRequest.identifier	Optional	Varchar	20		05.023.0012	
Order Date	MedicationRequest.authoredOn	Optional	G00.01, dd/mm/yyyy	10		05.023.0013	
Order Time	MedicationRequest.authoredOn	Optional	HH:MM:SS	8		05.023.0014	
Encounter ID	Encounter.identifier	Optional	Varchar	18		05.010.0001	
Episode ID	EpisodeOfCare.identifier	Optional	Varchar	50		05.009.0001	
Title (To be printed)							
Prescription Record							
Prescription ID	MedicationRequest.identifier	Optional	Varchar	20		05.023.0012	
Unique Individual Health Care Provider Number	Practitioner.identifier	Mandatory	Varchar	18		05.005.0001	
Health Care Provider Mobile Number	Practitioner.telecom	Optional	Char	10	Refer to Mobile Number (G00.06-02-05)	05.005.0007	
Unique Health Identification Number (UHID)	Patient.identifier	Conditional	Integer	12		G01.01	
Patient Information							
Provider's Patient ID	Patient.generalPractitioner	Mandatory	Varchar	18		05.003.0001	
Patient Name	Patient.name	Mandatory	Varchar	99	Refer to Name of the Person (G01.02)	05.003.0002	
Patient Age	Patient.birthDate	Mandatory	Age-year(s) (yyy) Integer (3) Age-Month(s) (mm) Integer(2)	7		05.003.0003	

Data Elements Labels	FHIR Label	Field Type	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
			Age-Day(s) (dd) Integer (2) Default Value: 999,99,99 no preceding zero [years, months, days]				
Patient Gender	Patient.Gender	0.*	Autofill	Varchar	Char M - Male F - Female, T – Transgender	Value set needs extension. The gender codes can be taken from metadata standards as that has been recommended by MeITY for all eGovernance and other applications. Suggested value sets are provided in column G.	
Patient Class	Encounter.class	Mandatory	Integer	2	CD05.047	05.003.0013	
Patient Address (05.003.0009)							
Patient Address Type	Address.use	Optional	Char	1		05.003.0010	
Premises Identifier	Address.line	Mandatory	Varchar	60		G02.03-00-02	House Number./DoorNumber/House Identifier/Flat Number Building Number./Plot Number Building Name/BuildingIdentifier
Sub Locality-1	Address.line	Optional	Varchar	60		G02.03-01-03	Block Name/Number or any other qualifier Street Number /Name/Mohalla/

Data Elements Labels	FHIR Label	Field Type	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
							Sector Number/any other qualifier
Locality	Address.line	Optional	Varchar	60		G02.03-03-03	
Country Code	Address.country	Optional	Integer	State - 2 District - 3 Sub-District - 5 Village - 6 Town - 6		G02.01	
District	Address.district	Mandatory	Integer	3	CD02.03	G02.01	
Sub-District	Address.line	Mandatory	Integer	5	CD02.04	G02.02-01	
Village	Address.line	Optional	Integer	6		G02.01	
Town	Address.city	Optional	Integer			CD02.06	
State	Address.state	Mandatory	Integer	2	CD02.02		
Pin	Address.postalCode	Mandatory	Integer	6		G02.04-01	
Name of country in English	Address.country	Mandatory	Varchar	50		G02.02-01	
Patient's email Address	Patient.telecom	Optional	Varchar	254	Refer to Email (G00.09)	G00.09	
Patient Mobile Number	Patient.telecom	Mandatory	Char	10	Refer to Mobile Number (G00.06-02-05)	05.003.0012	
AllergyIntolerance							
Allergy Product Code	AllergyIntolerance.code	Optional	Integer	5	CD05.018	05.018.0001	
Allergy Reaction Code	AllergyIntolerance.type	Optional	Varchar	10	https://www.hl7.org/fhir/valueset-allergy-intolerance-category.html	05.018.0003	
Allergy Severity Code	AllergyIntolerance.criticality	Optional	Integer	2	https://www.hl7.org/fhir/valueset-allergy-intolerance-criticality.html	05.018.0006	
Allergy Status	AllergyIntolerance.clinicalStatus	Optional	Integer	2	https://www.hl7.org/fhir/valueset-	05.018.0008	

Data Elements Labels	FHIR Label	Field Type	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
					allergyintolerance-verification.html		
Conditions							
Health Condition Code (Diagnosis)	Condition.code	Mandatory	Varchar	10	CD05.019/ICD10	05.020.0003	
Comorbidity Indicator	EpisodeOfCare.diagnosis.role	Optional	Integer	1		05.020.0008	
Comorbidity Health Condition Code	Condition.code	Optional	Varchar	10	CD05.019/ICD10	05.020.0009	
Pregnancy Indicator	Condition.code	Mandatory	Integer	1		05.003.0017	
Lab Orders							
Lab Order Code	ServiceRequest.code	Optional	Varchar	10	CD05.024	05.021.0022	
Lab Order Description	ServiceRequest.orderDetail	Optional	Varchar	50	NA		
Medication Request							
Drug Classification Code	MedicationKnowledge.code	Optional	Integer	2	NFI	05.023.0001	
Generic Drug Code	Medication.code	Mandatory	Integer	5	NFI	05.031.0004	
Brand Drug Code	Medication.code	Mandatory	Integer	10	Hospital specific	05.031.0006	
Brand Drug Name	Medication.code	Mandatory	Varchar	99	Hospital specific	05.031.0005	
Strength Value	Medication.ingredient.strength	Mandatory	Varchar	25		05.031.0011.	
Physical Form of Drug	Medication.form	Mandatory	Varchar	6	CD05.108	05.031.0010	
Dose	MedicationDispense.dosageInstruction	Mandatory	Varchar	60		05.023.0005	
Route of Administration	MedicationAdministration.dosage	Mandatory	Varchar	6	CD05.111	05.023.0002	
Medication Frequency	MedicationAdministration.dosage.rate[x]	Mandatory	Varchar	5	CD05.023	05.023.0003	
Medication Administration Interval	MedicationAdministration.dosage.rate[x]	Optional	Varchar	40		05.023.0004	
Body Site	MedicationAdministration.dosage.site	Optional	Integer	3	CD05.026	05.023.0007	
Contraindication	MedicationKnowledge.contraindication	Optional	Varchar	10	CD05.019	05.023.0018	
Medication Instructions	MedicationOrder.dosageInstruction	optional	Varchar	254		05.023.0024	
Medication Fills	MedicationRequest.dispenseRequest.number OfRepeatsAllowed	Optional	Integer	3		05.023.0019	
Fill No.	MedicationRequest.dispenseRequest.initialFill	Optional	Varchar	20		05.023.0027	

Data Elements Labels	FHIR Label	Field Type	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Quantity Ordered Value	MedicationDispense.quantity	Optional	Integer	10		05.023.0020	
Pharmacy Units	MedicationDispense.quantity	Optional	Varchar	25	CD05.109	05.023.0021	
Immunization Order (If applicable)							
Immunization Performer Identification Number	Immunization.performer	Optional	Integer	2		05.024.0004	
Immunization Product Code	Immunization.vaccineCode	Optional	Integer	3	CD05.036	05.024.0005	
Author's Detail (Prescriber details)							
Author Date	Composition.date	Mandatory	G00.01			05.019.0002	
Author Time	Composition.attester.time	Optional	HH:MM: SS	8		05.019.0001	
Author's Digital Signature	Signature.who	Mandatory					

eReferral eObject

An **eReferral** contains information about either a request for services to a higher level of care (an upward referral) OR the return of a patient to a lower level of care (a downward referral) for follow-up and continuing monitoring and care.

IRDA codes	Data Elements	Data Format	Maximum Size	Field Type	Value Set Codes	Values	Standard	Source	MDDS codes	Remark
Patient Demographics										
	Prescription ID	Varchar	20				MDDS		05.023.0012	
	Date of Prescription	dd/mm/yyyy	10				MDDS		G00.01	
	Patient's Name	Varchar	30	Auto filled			MDDS		G01.02-02	
	Alternate Unique Identification (UID) Type	Integer	2	Auto filled	CD05.007		MDDS		05.002.0001	

IRDA codes	Data Elements	Data Format	Maximum Size	Field Type	Value Set Codes	Values	Standard	Source	MDDS codes	Remark
	Patient's Unique Identification Number(UID)	Varchar	18	Auto filled			MDDS		05.002.0002	
	Gender	Char	1	Auto filled	G01.03		MDDS			
	Age	Integer	3	Auto filled			MDDS		05.003.0003	
	DOB	dd/mm/yyyy	10	Auto filled			MDDS			
	Patient Address	Varchar	99	Auto filled			MDDS		05.003.0009	
	Patient Mobile Number	Varchar	12	Auto filled			MDDS		05.003.0012	
	Email ID	Varchar	254	Auto filled			MDDS		G00.09	
	Pregnancy Indicator	Integer	1		1. Yes 2. No					
Referral Information										
	Referral Category Code	Char	1		CD05.067		MDDS		05.013.0009	
	Referral Category Name			Auto filled			MDDS		05.013.0018	
	Date of Referral	Refer to Date (G00.01)					MDDS		05.013.0010	
	Time of Referral	HH:MM:SS	8				MDDS		05.013.0011	
	Reason for Referral	Varchar	99				MDDS		05.013.0012	
	Patient Management Type	Integer	2		CD05.125		MDDS		05.013.0014	
	Cause of Mass Injury	Integer	2		CD05.066		MDDS		05.013.0007	
	Accident Location	Refer to Address of a Premises (G02.03)					MDDS		05.013.0008	
Facility Information										
	Referral Facility Identification Number	Integer	10		CD05.001		MDDS		05.008.0019	
	Referral Facility Type Code	Integer	2		CD05.002		MDDS		05.008.0020	
	Referral From Facility Identification Number	Integer	10		CD05.001		MDDS		05.008.0021	
	Referral From Facility Type Code	Integer	2		CD05.002		MDDS		05.008.0022	
Referring Provider Demographics										
	Provider UHID	Varchar	18				MDDS		05.005.0001	

IRDA codes	Data Elements	Data Format	Maximum Size	Field Type	Value Set Codes	Values	Standard	Source	MDDS codes	Remark
	Provider Type	char	1		CD05.120		MDDS		05.005.0002	
	Provider Name	Varchar	30				MDDS		05.005.0009	
	Provider Mobile	Varchar	12				MDDS		05.005.0007	
	Provider Role Code	Integer	2		CD05.009		MDDS		05.005.0010	
Attending Provider Demographics										
	Provider UHID	Varchar	18				MDDS		05.005.0001	
	Provider Type	char	1		CD05.120		MDDS		05.005.0002	
	Provider Name	Varchar	30				MDDS		05.005.0009	
	Provider Mobile	Varchar	12				MDDS		05.005.0007	
	Provider Role Code	Integer	2		CD05.009		MDDS		05.005.0010	
Referral Notes										
	Running e-referral Notes	Free text								

eEncounter eObject

An **eEncounter Note** contains summary information about each clinic visit and each hospital outpatient visit. It is defined as one patient receiving services from one provider on one specific date-of-service. In this case, the standard segment would contain patient identification, provider identification, facility identification, date-of-service, the reason for the encounter, today's diagnosis, and the patient disposition upon leaving.

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Header								

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Unique Health Identification Number (UHID)	Patient.identifier.value	1	Optional					This will be generated once MoHFW builds the master patient index for the country, as per NDHB we have to keep a place holder for the same. States can also build the state level patient or citizen registry and can use that ID here and later can roll up to the central.
Provider's Patient ID	Patient.identifier.value	0 to 1	Mandatory	Varchar	18		05.003.0001	The identifier used by a care provider (Individual or facility) to uniquely identify the patient.
Unique Facility Identification Number	Organization.identifier.value	0 to many	Mandatory	Integer	10	CD05.001	05.008.0001	Custodian of patient record or the object The government is working on creating a National Facility Registry for all the healthcare facilities (Public and Private) and will generate unique facility IDs for each. This ID will come through that registry. For now, states can use their facility ID if they have it at the state level that identifies each facility uniquely in the state. (Required to be used for PMJAY)
Unique Individual Health Care Provider Number	Practitioner.identifier.value	1	Mandatory	Varchar	18		05.005.0001	Similarly, as per NDHB, each healthcare provider like the doctor, nurse, etc will have a unique identifier that will be maintained Nationally in the provider registry. A placeholder for the same is required, till then medical council number or registration number of the doctor can be used here
Clinical Document Type Code	Composition.type.CodeableConcept.coding.code	1	Mandatory	Integer	2	CD05.046	05.019.0006	These will be backend values that will help to track all the clinical documents that will be generated by any application.
Document ID (Unique Eobject ID)	Composition.id (if it's object id it will be Composition .id , in case of business id it will be Composition.identifier)	1	Optional	Varchar	50		05.001.0023	Most of the application generates and store in their audit trail can just map with the MDDS code
Encounter Date	Encounter.participant.period	1	Mandatory	G00.01	10		05.010.0005	Every time a new visit is created for consultation or any other service system

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/val ue set	MDDS Label	Remark
Encounter ID	Encounter.identifier	1	Mandatory	Varchar	18		05.010.0001	automatically generates this to maintain the longitudinal record of a patient
Encounter Status	Encounter.status		Optional			https://www.hl7.org/fhir/valueset-encounter-status.html		
Medical record managing organization Name	Organization.name		Optional	String				The organization is the custodian of the patient record.
Title To be printed (On screen)								
Unique Health Identification Number (UHID)	patient.identifier.value	0 to many	Optional	Integer	12	G01.01		Nationally (State level) maintained patient's health ID
Provider's Patient ID	Patient.identifier.value	0 to many	Mandatory	Varchar	18		05.003.0001	Local Registration ID of the facility
Full Name in English	Patient.name	1	Mandatory	Refer to G01.02, Varchar	99		05.003.0002	Retrievable fields
Patient Age	Patient.birthDate	1	Mandatory	Age-year(s) (yyy) Integer(3) Age-Month(s) (mm) Integer(2) Age-Day(s) (dd) Integer (2) Default Value: 999,99,99 no preceding zero [years, months, days]	7		05.003.0003	
Gender Identification Code	Patient.gender	1	Mandatory	Char	1	M - Male F - Female T - Transgender	05.003.0021	
Patient Class	Patient.identifier	1	Mandatory	Integer	2	CD05.047	05.003.0013	This is used to categorize patients by the site where the encounter occurred, such as Emergency patients, Inpatients, Outpatients, etc.
Detailed Body								
Patient Details								

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Unique Health Identification Number (UHID)	Patient.identifier.value	0 to many	Optional	Integer	12	G01.01		Nationally (State level) maintained patient's health ID
Provider's Patient ID	Patient.identifier	0 to many	Mandatory	Varchar	18		05.003.0001	Local Registration ID of the facility
Full Name in English	Patient.name	1	Mandatory	Refer to G01.02, Varchar	99		05.003.0002	Retrievable fields
Patient Age	Patient.birthDate	1	Mandatory	Age-year(s) (yyy) Integer(3) Age-Month(s) (mm) Integer(2) Age-Day(s) (dd) Integer (2) Default Value: 999,99,99 no preceding zero [years, months, days]	7		05.003.0003	
Gender Identification Code	Patient.gender	1	Mandatory	Char	1	M - Male F - Female T - Transgender	05.003.0021	
Patient Class	Patient.identifier.value	1	Mandatory	Integer	2	CD05.047	05.003.0013	This is used to categorize patients by the site where the encounter occurred, such as Emergency patients, Inpatients, Outpatients, etc.
Patient Address (05.003.0009) -retrievable from patient registration								
Patient Address Type	Patient.address.type	1	Optional	Char	1	CD05.120	05.003.0010	
Premises Identifier	Patient.address.line or Patient.address.Text	1	Mandatory	Varchar	60		G02.03-00-02	House Number./DoorNumber/House Identifier/Flat Number Building Number./Plot Number Building Name/BuildingIdentifier
Sub Locality-1	Patient.address.line	1	Mandatory	Varchar	60		G02.03-01-03	Block Name/Number or any other qualifier Street Number /Name/Mohalla/ Sector Number/any other qualifier
Sub Locality-2	Patient.address.line	1	Optional	Varchar	60		G02.03-02-03	Major/Minor Landmark In Urban Land region (Town) or Rural Land region (Revenue village) for easy location of Premises

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/val ue set	MDDS Label	Remark
Locality	Patient.address.line or Patient.address.Text	1	Optional	Varchar	60		G02.03-03-03	Area Number/Area Name/Suburb/Sub district in case of Village/any other qualifier
District	Patient.address.district	1	Mandatory	Integer	3		G02.01	
Sub-District	Patient.address.line or Patient.address.Text	1	Optional	Integer	5	CD02.04	G02.02-01	
Village	Patient.address.line	1	Optional	Integer	6		G02.01	
Town	Patient.address.line or Patient.address.city	1	Optional	Integer		State-wise data		
State	Patient.address.line.state	1	Mandatory	Integer	2			
Pin	Patient.address.line.postal Code	1	Mandatory	Integer	6		G02.04-01	
Name of a Land region in English	Patient.address.country	1	Optional	Varchar	50		G02.02-01	Name of: Country / State / District / Sub District / Rural Land region (Revenue Village) / Urban Land region (Town) / any other Land identifier preferably in CAPITAL Letters
email		1	Optional	Varchar	254		G00.09	
Mobile Number		1	Mandatory	Char	10		05.003.0012	
Subjective Information								
Family History								
Family Member Medical History	FamilyMemberHistory.con dition	0....many	Optional	Varchar	4096		05.002.0024	
Family Member UID number	FamilyMemberHistory.iden tifier.value	0..many	Optional	G01.01			05.002.0025	
Family Member Relationship	FamilyMemberHistory.relat ionship	1	Mandatory	Integer	2	CD01.06	05.002.0027	
Health Condition Code (Family Member's)	FamilyMemberHistory.con dition.code	1..1	Mandatory	Varchar	10	ICD10/snom ed/ICD11, CD05.019	05.020.0003	Record only there is an active condition or a notifiable or NCD history
Family Member Age at Onset of Health Condition	FamilyMemberHistory.con dition.onset[x]		Optional	Age-year(s) (yyy) Integer(3) Age-Month(s)				

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/val ue set	MDDS Label	Remark
				(mm) Integer(2) Age-Day(s) (dd) Integer (2) 999,99,99 no preceding zero [years, months, days]				
Health Condition status	FamilyMemberHistory.stat us	1	Mandatory	Integer	2	CD05.021	05.020.0007	
Family Member Cause of Death Known Indicator	FamilyMemberHistory.con dition.contributedToDeath	0..1	Optional	Integer	1	Values: 0- No 1- Yes The default value will be 0.	05.019.0011	
Family Member Age At Death	FamilyMemberHistory.dec eased[x]	0..1	Optional	Age-year(s) (yyy) Integer(3) Age-Month(s) (mm) Integer(2) Age-Day(s) (dd) Integer (2) 999,99,99 no preceding zero [years, months, days]			05.019.0012	
Attachments								
Provider Details								
Unique Facility Identification Number	Organization.identifier.valu e	0 to many	Mandatory	Integer	10	CD05.001	05.008.0001	Custodian of patient record or the object The government is working on creating a National Facility Registry for all the healthcare facilities (Public and Private) and will generate unique facility IDs for each. This ID will come through that registry. For now, states can use their facility ID if they have at the state level that identifies each

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/val ue set	MDDS Label	Remark
								facility uniquely in the state. (Required to be used for PMJAY)
Unique Individual Health Care Provider Number	Practitioner.identifier.value	1	Mandatory	Varchar	18		05.005.0001	Similarly, as per NDHB, each healthcare provider like doctor, nurse, etc will have a unique identifier that will be maintained Nationally in the provider registry. A placeholder for the same is required, till then medical council number or registration number of the doctor can be used here
Visit Details								
Encounter Date	Encounter.participant.period	1	Mandatory	dd/mm/yyyy	10		05.010.0005	Every time a new visit is created for consultation or any other service system automatically generates this to maintain the longitudinal record of a patient
Encounter ID	Encounter.identifier.value	1	Mandatory	Varchar	18		05.010.0001	
Encounter Type	Encounter.type	1 to many	Mandatory	Integer	2	CD05.047	05.010.0002	
Episode of care	Encounter.episodeOfCare	0..*	Optional					
Episode ID	this info will come from the Episode of Care resource linked as a reference in Encounter.episode of care	1	Optional	Varchar	50		05.009.0001	Automatically generated by the system and NSF card should store this information for the encounter info they store
Episode Type	this info will come from the Episode of Care resource linked as reference in Encounter.episodeOfCare	1	Optional	Integer	1	1 - New 2 - Ongoing 3 - Active 4 - Inactive	05.009.0002	
Episode Status	this info will come from Episode of Care resource linked as refrence in Encounter.episodeOfCare	1	Optional			planned Waitlist Active Onhold Finished Cancelled Entered in error	05.008.0010	
Facility Specialty Code	HealthcareService.specialty	0 to many	Optional	Integer	3	CD05.011	05.008.0010	
Clinical Details								
Patient's Clinical History								

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Existing/comorbidity Health Condition Code	Condition.code	0 to many	Mandatory	Varchar	10	ICD10/11/SNOMED	05.020.0003	
Health Condition status	Condition.clinicalStatus	1	Mandatory			CD05.021	05.020.0007	For the next encounter, all the cured and closed diagnosis will be shown here with a logic built by the healthcare facility (eg: till what date a closed diagnosis should be shown)
Past Health Condition Onset Date	Condition.onset[x]	1	Optional	Refer to Date (G00.01)	10		NA	
Chief Complaints								
Chief Complaint ID	Condition.code	0 to many	Mandatory	Varchar			NA	Can have more than 1 cardinality)
Chief Complaint Name	Condition.code	0 to many	Mandatory	Varchar		ICD 10 (Signs & Symptoms)	NA	
Body Site	Condition.bodySite	1	Mandatory	Integer	3	CD05.026	05.023.0007	
Duration	Condition.onset[x]	1	Mandatory	Integer			NA	
Patient Allergies								
Allergy Product Code	AllergyIntolerance.code.coding.code	0 to many	Mandatory	Integer	5	CD05.018	05.018.0001	This information is collected during the first consultation, but the treating doctor can add more allergies if reported by the patient or through an adverse event reported by the hospital in the future.
Allergy Product Description	AllergyIntolerance.reaction.substance.codeableConcept.display	0 to many	Optional	Varchar	99		05.018.0002	
Allergy Reaction Code	AllergyIntolerance.reaction.substance.codeableConcept.code.code	0 to many	Optional	Varchar	10	ICD10	05.018.0003	
Allergy Reaction Name	AllergyIntolerance.reaction.substance.codeableConcept.code.display	0 to many	Optional	Varchar	99	ICD10		
Allergy Severity Code	AllergyIntolerance.criticality	0..1	Optional	Integer	2	CD05.020	05.018.0006	
Adverse Event Type	AllergyIntolerance.reaction.manifestation			Varchar	10	CD05.019		
Allergy Status	AllergyIntolerance.clinicalStatus	0 to many	Mandatory	Integer	2	CD05.021	05.018.0008	
Observations								

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/val ue set	MDDS Label	Remark
Vitals https://www.hl7.org/fhir/observation-vitalsigns.html								
Vital Sign Result Time	Observation.effective[x]	0 to many	Optional	HH:MM:SS	8		05.017.0001	
Vital Sign Result Type	Observation.category	0 to many	Optional	Integer	2	CD05.041/ https://www .hl7.org/fhir/ observation- vitalsigns.ht ml/	05.017.0002	
Vital Signs Result Status	Observation.status	0 to many	Mandatory	Integer	2	CD05.038	05.017.0003	Can be configured as per the specialty or clinical problem an application is trying to solve. A facility or department can decide which vitals they would like to capture. The vital parameters are provided in the associated code directory which is provided in another excel in this folder.
Vital Sign Result Value	Observation.value[x]	0 to many	Optional	Varchar	20		05.017.0004	
Vital Sign Result Unit	Observation.value[x]	0 to many		Integer	2	CD05.025	05.017.0005	
Vital Sign Result Interpretation	Observation.interpretation	0 to many	Optional	Integer	2	CD05.135	05.017.0006	
Vital Sign Result Reference Range - lower limit	Observation.referenceRange.low	0 to many	Optional	Integer	3	CD05.039	05.017.0007	
Vital Sign Result Reference Range - Upper limit	Observation.referenceRange.high	0 to many	Optional	Integer	3	CD05.039	05.017.0008	Each vital parameter will have cardinality 1 and is captured in every encounter.
Vital Sign Result Date	Observation.issued	0 to many	Optional	dd/mm/yyyy	10		05.017.0009	
Vital Sign Result ID	Observation.identifier.value	0 to many	Optional	Integer	2		05.017.0010	
Examination								
Examination Type	Observation.category	0 to many	Optional	Integer	3	CD05.061	05.016.0001	There will be specific templates for each Type as per the medical specialty.
Examination Finding	Observation.code	0 to many	Mandatory	Varchar	4096		05.016.0002	
Examined System	Observation.code	0 to many	Optional	Integer	2	CD05.033	05.016.0003	
Assessment								
Diagnosis								
Health Condition Type	Condition.category	1 to many	Mandatory	Integer	3	CD05.022	05.020.0001	
Health Condition name	Condition.code..coding.display	1 to many	Mandatory	Varchar	99	CD05.019	05.020.0002	
Health Condition Code	Condition.code..coding.code	1 to many	Mandatory	Varchar	10	CD05.019 (ICD10/SNO MeD)	05.020.0003	Diagnosis ID

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Health Condition Category	Condition.category	1 to many	Optional	Char	1	Valueset: S-Suspected, P-Presumptive L-lab confirmed C-Clinically confirmed	05.020.0005	
Diagnosis Priority	Encounter.diagnosis.rank	1 to many	Optional	Integer	1	Valueset: 1-Primary 2-Secondary	05.020.0006	
Present Health Condition Onset Date	Condition.onset[x]	1 to many	Optional	Refer to Date (G00.01), dd/mm/yyyy	10		05.020.0010	Auto captures date of entry
Health Condition Status	Condition.clinicalStatus	1 to many	Optional	Integer	2	CD05.021	05.020.0007	System should facilitate closure of a cured condition
Comorbidity Indicator	EpisodeOfCare.diagnosis.role	1 to many	Optional	Integer	1		05.020.0008	
Care Plan Details (Clinical Order)								
Treatment Plan details (If applicable)								
Proposed line of treatment	CarePlan.activity	0 to many	Optional	Integer	5	CD05.028	05.007.0038	Once we have standard treatment guidelines those can be used or the facility may have defined their own set of treatment plan Note: Until STGs arrives we can use package IDs for insurance beneficiary (With cardinality if there are more than one package applicable in case of multiple surgeries during the same patient stay or episode (two open episodes))
Treatment Plan code	CarePlan.code.codeableConcept.code	0 to many	Optional	varchar	10		NA	
Treatment Plan Name	CarePlan.code.codeableConcept.display	0 to 1	Optional	Varchar	120		NA	
Package Details (If Applicable)								
Package Item Name	List.code.codeableConcept.display	0 to many	Optional	Varchar	99		05.007.0020	Package refers to a group of service items with sub-limits.

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/val ue set	MDDS Label	Remark
Package Code	List.code.codeableConcept. code	0 to many	Optional	Varchar	12		NA	Applicable to health insurance treatment packages
Order Info (Applicable for all orders)								
Order Date	CarePlan.created	1	Mandatory	Refer to Date (G00.01)	10		05.023.0013	
Order Time	CarePlan.period	1	Mandatory	HH:MM:SS	8		05.023.0014	
Order Group ID	CarePlan.identifier.value	1	Optional	Varchar	10		05.025.0007	Applicable for composite orders or order set (since order sets are used in janta clinic flow) example annual health and wellness check-up
Order ID	CarePlan.identifier.value	1	Optional	Varchar	12		05.025.0004	
Order Status	CarePlan.status	1	Mandatory	Char	2	CD05.121	05.025.0008	
Lab Investigations								
Lab Order Code	ServiceRequest.code	0 to many	Mandatory	Varchar	10	CD05.024/LO INC	05.021.0022	
Lab Order Description	DiagnosticReport.category	0 to many	Optional	Varchar	50		NA	
Lab Result ID	DiagnosticReport.identifier. value	0 to many	Optional	Varchar	10		05.021.0025	These values will be visible only when there is a follow-up visit for result awaited or a follow up visit for the same episode
Result Status	DiagnosticReport.status	0 to many	Mandatory	Char	2	CD05.038	05.021.0004	
Result Value	DiagnosticReport.result	0 to many	Optional	Varchar	20		05.021.0005	
Result Interpretation	DiagnosticReport.conclusio n	0 to many	Optional	Integer	2	CD05.135	05.021.0006	
Result Reference Range - lower limit	Observation.referenceRang e.low	0 to many	Optional	Integer	7	CD05.039	05.021.0007	
Result Reference Range - Upper limit	Observation.referenceRang e.high	0 to many	Optional	Integer	7	CD05.039	05.021.0008	
Radiology Investigations								
Radiology Procedure Code	Procedure.codeableConcep t.coding.code	0 to many	Optional	Varchar	18	CD05.043	05.022.0008	
Radiology Procedure Name	Procedure.codeableConcep t.coding.display	0 to many	Optional	Varchar	255	CD05.043	05.022.0007	
Radiology Result Status	DiagnosticReport.status	0 to many	Mandatory	Integer	2	CD05.038	05.022.0009	Applicable to follow-up visit
Radiology Result ID	DiagnosticReport.identifier. value	0 to many	Optional	Varchar	10		05.022.0010	

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/val ue set	MDDS Label	Remark
scanned report attachment	DocumentReference.conen t.attachement	0 to many						
Non-radiology Procedure Orders								
Procedure Code	Procedure.codeableConcep t.coding.code	0 to many	Mandatory	Varchar	10	CD05.043	05.026.0003	If applicable
Procedure Name	Procedure.codeableConcep t.coding.display	0 to many	Optional	Varchar	255	CD05.043	05.026.0001	
Rx Orders								
Prescription ID	MedicationRequest.identifi er.value	1	Mandatory	Varchar	20		05.023.0012	
Generic Drug Code	Medication.codeableConce pt.coding.code	0 to many	Mandatory	Integer	5	CD05.104	05.031.0004	
Brand Drug Code	MedicationRequest.medica tionCodeableConcept.codi ng.code	0 to many	Optional	Integer	10	CD05.105	05.031.0006	
Brand Drug Name	MedicationRequest.medica tionCodeableConcept.codi ng.code.display	0 to many	Mandatory	Varchar	99	CD05.105	05.031.0005	
Strength Value	MedicationRequest.medica tionCodeableConcept.codi ng.code	0 to many	Optional	Varchar	25		05.031.0011	
Route of Administration	MedicationRequest.dosage Instruction.route.coding.co de	0 to many	Optional	Varchar	6	CD05.111	05.023.0002	
Medication Frequency	MedicationRequest.dosage Instruction.additionalInstru ction.coding	0 to many	Optional	Varchar	5	CD05.023	05.023.0003	
Medication Administration Interval	MedicationRequest.dosage Instruction.additionalInstru ction.coding	0 to many	Optional	Varchar	40		05.023.0004	
Dose	MedicationRequest.medica tionCodeableConcept.codi ng.code	0 to many	Optional	Varchar	60		05.023.0005	
Medication Stopped Indicator	MedicationRequest.dosage Instruction.additionalInstru ction.coding	0 to many	Mandatory	Integer	1			
Medication Status		0 to many	Optional	Integer	2	CD05.123	05.023.0010	

Data Elements Labels	Updated FHIR Label	Cardinality	Field Type	Data Format (String, Varchar, Integer, Value set)	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
Medication Fills	MedicationRequest.dispenseRequest.initialFill	0 to many	Optional	Integer	3		05.023.0019	
Medication Fill No.	MedicationRequest.dispenseRequest.numberOfRepeatsAllowed	0 to many	Optional	Varchar	20		NA	
Quantity Ordered Value	MedicationDispense.dispenseRequest.initialFill.quantity	0 to many	Optional	Integer	10		05.023.0020	
Pharmacy Units	MedicationRequest.dispenseRequest.performer	0 to many	Optional	Varchar	25	CD05.109	05.023.0021	
Immunization Order (If applicable)								
Immunization Performer Identification Number	Immunization.performer	0 to many	Optional	Varchar	18		05.024.0004	
Immunization Product Code	Immunization.vaccineCode	0 to many	Mandatory	Integer	3	CD05.036	05.024.0005	
Medication Series No.	Immunization.protocolApplied.seriesDoses[x]	0 to many	Optional	Integer	2		05.024.0003	
Immunization Administered Date	Immunization.occurrence[x]	0 to many	Mandatory	G00.01, dd/mm/yyyy			05.024.0008	
Follow Up Order								
Follow up date	Appointment.start	0 to many	Optional	G00.01,dd/mm/yyyy	10		NA	
Follow up interval	Appointment.slot	0 to many	Optional				NA	
Patient Instruction	Appointment.comment	0 to many	Optional	Free Text			NA	
Author Details								
Author Date	Composition.date	1	Mandatory	G00.01,dd/mm/yyyy	10		05.019.0002	Auto captured with role-based access control/ Doctor's digital signature who created the encounter note
Author Time	Composition.attester.time	1	Optional	HH:MM:SS	8		05.019.0001	
Author's Digital Signature	Signature.who	1	Mandatory					

eDiagnostic eObject: Laboratory Template

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Header (Encrypted and stored) This data will be machine readable and not viewed on the screen								
Unique Health Identification Number (UHID)	Patient.identifier	1	Optional	Bits	16	CD05.001	05.008.0025	India still doesn't have a Master Patient Index; hence this field can be kept optional for now. But for vendors who handle Ayushman Bharat patients or any state health scheme or a state which has identified each patient uniquely (Kerala) the member ID/State allocated patient ID can be utilized for such patients. E.g.: PMJAY ID, BHAMASHA ID, Kerala Patient Health ID=ADHAAR
Unique identification (UID)	person.identifier	0 to 1	Optional	Integer	12		G01.01	To be used for ADHAR (UIDAI) if authorized, This is used for person identification.
Provider's Patient ID	Patient.identifier	1 to many	Mandatory	Varchar	18		05.003.0001	This will be generated once MoHFW builds the master patient index for the country, as per NDHB we have to keep a place holder for the same. States can also build the state level patient or citizen registry and can use that ID here and later can roll up to the central.
Alternate Unique Identification Number (UID) Type	Identifier.type	0 to many	Optional	Integer	2	CD05.007	05.002.0001	
Alternate Unique Identification Number (UID)	Person.identifier	0 to many	Mandatory	Varchar	Max.Size=18 10-PAN Card08 - 08- PassportNo. 18 -VoterID 18 -Any other Identifier		05.002.0002	Till the time the UHID is unavailable, patient alternate id can be used in place of a unique identifier. For a hospital setting it can be the patient's AADHAR number. and for those enrolled in vertical programs, the IDs allotted to each patient under that program can be utilized for example, NIKSHAY ID. This can be extended to accommodate identifiers from various vertical programs for the same patient until we have UHID and a National Patient Master Index.
Unique Individual Health Care Provider Number	Practitioner.identifier	1	Mandatory	Varchar	18		05.005.0001	Similarly as per NDHB each healthcare provider like doctor, nurse etc will have a unique identifier that will be maintained Nationally in the provider registry. A placeholder for the same is required, till then medical council number or registration number of the doctor can be used here
Facility Type Code	Organization.type	0 to many	Optional	Integer	2	CD05.002	05.008.0002	Lab, radiology centre etc

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Facility Global Unique Identifier (GUID)	Patient.managingOrganization	1	Optional	Bits	16	CD05.001	05.008.0025	Custodian of patient record or the object Government is working on creating a National Facility Registry for all the healthcare facilities (Public and Private) and will generate unique facility IDs for each. This ID will come through that registry. For now, states can use their own facility ID if they have at state level that identifies each facility uniquely in the state. (Required to be used for PMJAY)
Lab ID	Patient.managingOrganization	1	Optional	Integer	10		05.021.0023	It is an identifier for the local Labs. This identifier will come from the application-level Lab Master. This has to be used in case the Lab identifier cannot be taken from facility master Index which in in works for now.
Lab Order Code	ServiceRequest.code		Mandatory	Varchar	10	CD05.024:LOINC	05.021.0022	The order code for the requested observation, test, and/or battery. Note: This can be based on local and/or standardized order codes. E.g., For Laboratory Orders, refer to LOINC code directory
Result category	DiagnosticReport.category	0 to many	Mandatory	Varchar	10	01- Vital Signs 02- Laboratory Results 03- Radiology Results 04- Blood Bank Screening CD05.040	05.021.0009	
Clinical Document Type Code	Composition.type	1	Mandatory	Integer	2	CD05.046 + E-prescription Object E-Encounter Object E-Discharge summary E-Diagnostic Report	05.019.0006	These will be backend values that will help to track all the clinical documents that will be generated by any application.
Document ID/Object ID	Composition.type	1	Optional	Varchar	50		05.001.0023	
Result Date	Encounter.identifier	1	Mandatory	Date-DD/MM/YYYY	10		05.021.0001	

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Result Time	Encounter.appointment	1	Optional	HH:MM: SS	8		05.021.0002	
Detailed Body								
Title To be printed								
UHID (Unique Health Identification Number)	Patient.identifier	0 to many	Optional	Integer	12	G01.01		
Provider Patient ID	Patient.identifier	0 to many	Mandatory				05.003.0001	Local registration ID used by facilities
Full Name in English	Patient.name	1	Optional	Varchar	99		05.003.0002	Retrievable fields
Patient Age	Patient.birthDate	1	Optional	Age-year(s) (yyy) Integer (3) Age-Month(s) (mm) Integer(2) Age-Day(s) (dd) Integer (2) Default Value: 999,99,99 no preceding zero [years, months, days]	7		05.003.0003	
Patient Gender	Patient.gender	1	Optional	Char	1	M - Male F - Female T - Transgender	G01.03	
Patient Mobile Number	Patient.telecom (ContactPoint.value)	1	Optional	Char	10		05.003.0012	
Lab ID	Patient.managingOrganization	1	Optional	Integer	10		05.021.0023	It is an identifier for the local Labs. This identifier will come from the application-level Lab Master. This must be used in case the Lab identifier cannot be taken from facility master Index which in in works for now.
Lab Name		1	Optional	Char	99			
Lab Type		1 to many	Optional	Integer	1	Values : 1- Clinical Pathology 2- Clinical Microbiology	05.021.0024	To further sub categorize the lab type

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
						3- Clinical Biochemistry		
Lab Order Code	ServiceRequest.code		Mandatory	Varchar	10	CD05.024	05.021.0022	The order code for the requested observation, test, and/or battery. Note: This can be based on local and/or standardized order codes. E.g., For Laboratory Orders, refer to LOINC code directory
Lab Report Details								
Lab Result ID	DiagnosticReport.identifier	0 to many	Optional	Varchar	10		05.021.0025	
Specimen Type	DiagnosticReport.Specimen							
Result Date	Encounter.identifier	1	Mandatory	Date- DD/MM/YYYY	10		05.021.0001	
Result Time	Encounter.appointment	1	Optional	HH:MM:SS	8		05.021.0002	
Result Type		0 to1	Optional			LOINC Codes for result type	05.021.0003	
Result Method		0-Many	Optional	Varchar	100			This is a field to capture method used for processing of sample. Currently has to be added as a new code directory to MDDS code directories but can be from local values in the short term.
Result Status	DiagnosticReport.status	0 to1		Char	2	CD05.038	05.021.0004	
Result Value	DiagnosticReport.result	0 to1		Varchar	20		05.021.0005	The value of the result
Unit of Measurement				Varchar	20	Units of measurement CD05.025		units of measure, if applicable.
Result Interpretation	DiagnosticReport.conclusion	0 to many	Mandatory	Integer	2	CD05.135	05.021.0006	
Result Reference Range - lower limit	Observation.referenceRange.low	0 to many	Optional	Integer/varchar	7/50		05.021.0007	
Result Reference Range - Upper limit	Observation.referenceRange.high	0 to many	Optional	Integer/varchar	7/50		05.021.0008	
Diagnostic report interpreter		0 to many	Optional					
Attachment		0 to many	Optional	Binary				
Author Details								

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Author Date	Composition.date	1	Mandatory	dd/mm/yyyy	10	G00.01	05.019.0002	Auto captured with role-based access control/ Doctor's digital signature who created the encounter note
Author Time	Composition.attester.time	1	Optional	HH:MM:SS	8		05.019.0001	
Author's Digital Signature	Signature.who	1	Mandatory					

eDiagnostic eObject : Radiology Template

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Header (Encrypted and stored) This data will be machine readable and not viewed on the screen								
Unique Health Identification Number (UHID)	Patient.identifier	1	Optional	Bits	16	CD05.001	05.008.0025	India still doesn't have a Patient Master Index; hence this field can be kept optional for now. But for vendors who handle Ayushman Bharat patients or any state health scheme or a state which has identified each patient uniquely (Kerala) the member ID/State allocated patient ID can be utilized for such patients. E.g.: PMJAY ID, BHAMASHA ID, Kerala Patient Health ID=ADHAAR
Unique identification (UID)	person.identifier	0 to 1	Optional	Integer	12		G01.01	To be used for ADHAR (UIDAI) if authorized, this is used for person identification.
Provider's Patient ID	Patient.identifier	1 to many	Mandatory	Varchar	18		05.003.0001	This will be generated once MoHFW builds the mater patient index for the country, as per NDHB we must keep a place holder for the same. States can also build the state lever patient or citizen registry and can use that ID here and later can roll up to the central.

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Alternate Unique Identification Number (UID) Type	Identifier.type	0 to many	Optional	Integer	2	CD05.007	05.002.0001	
Alternate Unique Identification Number (UID)	Person.identifier	0 to many	Mandatory	Varchar	Max.Size=18 10-PAN Card08 - 08- PassportNo. 18 -VoterID 18 -Any other Identifier		05.002.0002	Till the time the UHID is unavailable, patient alternate id can be used in place of a unique identifier. For a hospital setting it can be the patient's AADHAR number. and for those enrolled in vertical programs, the IDs allotted to each patient under that program can be utilized for example, NIKSHAY ID. This can be extended to accommodate identifiers from various vertical programs for the same patient until we have UHID and a National Patient Master Index.
Unique Individual Health Care Provider Number	Practitioner.identifier	1	Mandatory	Varchar	18		05.005.0001	Similarly, as per NDHB each healthcare provider like doctor, nurse etc will have a unique identifier that will be maintained Nationally in the provider registry. A placeholder for the same is required, till then medical council number or registration number of the doctor can be used here
Facility Type Code	Organization.type		Optional	Integer	2	CD05.002	05.008.0002	Lab, radiology centre etc
Facility Global Unique Identifier (GUID)	Patient.managingOrganization	1	Optional	Bits	16	CD05.001	05.008.0025	Custodian of patient record or the object Government is working on creating a National Facility Registry for all the healthcare facilities (Public and Private) and will generate unique facility IDs for each. This ID will come through that registry. For now, states can use their own facility ID if they have at state level that identifies each facility uniquely in the state. (Required to be used for PMJAY)
Radiology Center ID	Patient.managingOrganization	1	Mandatory	Integer	10	https://www.aerb.gov.in/images/PDF/NuclearMedicine/rsd1feb2018.pdf	05.022.0001	It is an identifier for the local radiology diagnostic centres, assigned by BARC. This identifier will come from the application-level Radiology Center Master.
Radiology Center Type		1	Optional	Integer	1		05.022.0002	

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Result category	DiagnosticReport.category		Mandatory	Varchar	10	01- Vital Signs 02- Laboratory Results 03- Radiology Results 04- Blood Bank Screening	05.021.0009	
Clinical Document Type Code	Composition.type	1	Mandatory	Integer	2	CD05.046 + E-prescription Object E-Encounter Object E-Discharge summary E-Diagnostic Report	05.019.0006	These will be backend values that will help to track all the clinical documents that will be generated by any application.
Document ID/Object ID	Composition.type	1	Optional	Varchar	50		05.001.0023	
Radiology Procedure Date	Encounter.identifier	1	Mandatory	Date-DD/MM/YYYY			05.022.0003	
Radiology Procedure Time	Encounter.appointment	1	Optional	HH:MM: SS	8		05.022.0004	
Radiology Investigation Code								
Detailed Body								
Title To be printed								
UHID (Unique Health Identification Number)	Patient.identifier	0 to many	Optional	Integer	12	G01.01		
Provider Patient ID	Patient.identifier	0 to many	Mandatory				05.003.0001	Local registration ID used by facilities
Full Name in English	Patient.name	1	Optional	Varchar	99		05.003.0002	Retrievable fields
Patient Age	Patient.birthDate	1	Optional	Age-year(s) (yyy) Integer (3) Age-Month(s) (mm) Integer(2) Age-Day(s) (dd) Integer (2)	7		05.003.0003	

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
				Default Value: 999,99,99 no preceding zero [years, months, days]				
Patient Gender	Patient.gender	1	Optional	Char	1	M - Male F - Female T - Transgender	G01.03	
Patient Mobile Number	Patient.telecom (ContactPoint.value)	1	Optional	Char	10		05.003.0012	
Facility ID	Patient.managingOrganization	1	Optional	Integer	10		05.021.0023	It is an identifier for the local Labs. This identifier will come from the application-level Lab Master. This has to be used in case the Lab identifier cannot be taken from facility master Index which in in works for now.
Facility Name								
Facility Type		1 to many	Optional	Integer	1		05.021.0024	To further sub categorize the lab type
Radiology Procedure Code	Procedure.code		Optional	Varchar		LOINC	05.022.0008	
Radiology Procedure Name	Procedure.code		Mandatory	Varchar	10	LOINC/local name	05.022.0007	The order code for the requested observation, test, and/or battery.
Rad Report Details								
Radiology Result ID	DiagnosticReport.identifier	0 to many	Optional	Varchar	10		05.022.0010	
Result Date	Encounter.identifier	1	Mandatory	Date-DD/MM/YYYY			05.021.0001	
Result Time	Encounter.appointment	1	Optional	HH:MM:SS	8		05.021.0002	
Result Type		0 to1	Optional			LOINC Codes	05.021.0003	
Result Status	DiagnosticReport.status	0 to1		Char	2	CD05.038	05.021.0004	
Result Interpretation	DiagnosticReport.conclusion	0 to many	Mandatory	Integer	2		05.021.0006	
Diagnostic report interpreter		0 to many	Optional					
Report Attachment		0 to many	Optional	Binary				

Data Elements Label	FHIR label	Cardinality	Field type	Data Format (String, Varchar, Integer, Value set)	Maximum size	Applicable code directory	MDDS label	Remarks
Key Image Attachment		0 to many	Optional	Binary				
Image Attachment		0 to Many	Optional	Binary				
Author Details								
Author Date	Composition.date	1	Mandatory	dd/mm/yyyy	10	G00.01	05.019.0002	Auto captured with role-based access control/ Doctor's digital signature who created the encounter note
Author Time	Composition.attester.time	1	Optional	HH:MM:SS	8		05.019.0001	
Author's Digital Signature	Signature.who	1	Mandatory					

eDischarge eObject

An **eDischarge Summary** contains summary information about each inpatient hospital stay. The eDischarge Summary is perhaps the most crucial object of all. It serves an important role in linking episodes of care thus preserving longitudinal continuity-of-care. It serves as an attachment to any eReferral between venues of care. It serves as a summary of services which can help justify the services declared on an eClaim. Along with the eClaim, it serves other epidemiological and administrative needs as well in helping to calculate burden-of-disease, and quality and utilization measures. Well, the uses of a reliable, standardized eDischarge Summary are nearly endless.

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
eDischarge Object Header DischargesummaryHeader FHIR resource Type= "Bundle" type ="Document" Resource Identifier LOINC Code - 28655-9 (Attending Discharge Summary) Last Update: "2013-05-28T22:12:21Z"								
Discharge Summary- Composition Details (Follow NRCeS specification)								
eDischarge Object Header								
Unique eDischarge Object ID	document.identifier	1..1	System Generat ed	String	50			
Patient's UHID	Patient.identifier.id Reference- Patient Resource- <a href="https://www.nrces.in/n
dhm/fhir/r4/StructureD
efinition-Patient-
definitions.html#Patien
t">https://www.nrces.in/n dhm/fhir/r4/StructureD efinition-Patient- definitions.html#Patien t	0..1	Mandat ory- Govern ment Program s & shemes Optional - Private	String				

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
			Providers					
Provider Patient ID	Patient.identifier.id	1..1	Autofill	String				
Local Provider's/Hospital's Patient ID	Patient.identifier.id Reference- Patient Resource- https://www.ncres.in/ndhm/fhir/r4/StructureDefinition-Patient-definitions.html#Patient	1..1	Mandatory- Should be autofilled if Hospital has an EMR	String				
Unique Facility Identification Number	Organization.identifier		Autofill	Integer	18		Can be autofill as per the user log in	
Treating Physician ID	Patient.generalPractitioner		Mandatory	Integer	20		reference = participantRole where participantRole.practitioner.Identifier is mapped to Treatment/Physician ID and participantRole.code = "treatment doctor"	
Patient Detail Section Reference Resource- https://www.ncres.in/ndhm/fhir/r4/StructureDefinition-Patient-definitions.html#Patient								
Patient's UHID	Patient.identifier.id	0..1	Mandatory- Government Programs & schemes	String				

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
			Optional - Private Providers					
Provider Patient ID	Patient.identifier.id	1..1	Autofill	String				
Prefix	Patient.name.prefix	0..1	Optional					
Patient Name	Patient.name Patient.name.text	1..1	Autofill	Varchar	99			
Patient Family Name	Patient.name.family	1..1	Autofill	Varchar				
Patient Full Name	Patient.name.given	1..1	Autofill	Varchar				
	Patient.name.use	0..1	Optional	Varchar		http://hl7.org/fhir/R4/valueset-name-use.html		
Telecom Resource								
Patient's email Address	Patient.telecom.value	0..*	Autofill	Varchar	254			
Mobile number	Patient.telecom.value	0..*	Autofill	Char	10			
Patient Gender	Patient.gender	1..1	Autofill	Code		Char M - Male F - Female T - Transgender		Value set needs extension. The gender codes can be taken from meta data standards as that has been recommended by MeITY for all eGovernance and other applications. Suggested value sets are provided in column G.
Date of Birth	Patient.birthDate	0..1	Autofill	Date (DDMM YYYY)				FHIR Date Format and Indian date format is different. India uses DDMMYYYY hence this element format needs an extension.

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Patient Age	No corresponding resource found in FHIR	1..1	Autofill	Age- year(s) (yyy) Integer(3) Age- Month(s) (mm) Integer(2) Age- Day(s) (dd) Integer(2)	7		Age is to be automatically calculated if Date of birth is entered/available; and vice versa. For this, the age will be approximated with the assumption that the patient was born on the 1st day of January of the year that the entered age appears to point to. The record display will need to clearly show that this age is an approximated one.	The element is missing in FHIR, an extension for age is recommended, as in India most of the patients do not remember their DOB but remembers the approximate age.
Patient Address	Patient.address	0..*	Autofill					
Patient Address - Patient.address Reference Resource- http://hl7.org/fhir/R4/datatypes-definitions.html#Address								
Address Use	Address.use	0..1	Optional	Code		http://hl7.org/fhir/R4/valueset-address-use.html		
Address Type	Address.type	0..1	Optional	Code		http://hl7.org/fhir/R4/valueset-address-type.html		

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Premises Identifier House Number./Door Number/House Identifier/Flat Number Building Number./Plot Number Building Name/Building Identifier	Address.line	0..*	Autofill	String				
Sub Locality-1 "Block Name/Number or any other qualifier Street Number /Name/Mohalla/ Sector Number/any other qualifier"	Address.line	0..*	Optional	String	50			
Sub Locality-2 Major/Minor Land mark In Urban Land region (Town) or Rural Land region (Revenue village) for easy location of Premises	Address.line	0..*	Optional	String				
Locality Area Number/Area Name/Suburb/Su	Address.line	0..*	Optional	Varchar	50			

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
b district in case of Village/any other qualifier								
District	Address.district	0..1	Autofill	Code	3	District Codes https://censusindia.gov.in/Census_Data_2001/PLCN/plcn.html		These two elements need extension as it is recommended to use district and subdistrict codes published by RGI to standardize patient addresses. The link for the same is provided under column G. This is important to enable countrywide interoperability and health delivery portability
Village		0..1	Autofill	Code	6	Village Codes https://censusindia.gov.in/Census_Data_2001/PLCN/plcn.html		The village element is specific to India as most of our population lives in Rural India, hence the minimum specification should support capturing village details to enable multiple health programs under NRHM,
Town	Address.city	0..1	Autofill	Code		Town Codes https://censusindia.gov.in/Census_Data_2001/PLCN/plcn.html		It is recommended to use City and state codes published by RGI. The link for the same is provided under column G. This is important to enable countrywide interoperability and health delivery portability.
State	Address.state	0..1	Autofill	Code	2	State Codes https://censusindia.gov.in/Census_Data_2001/PLCN/plcn.html		
Pin	Address.postalCode	0..1	Autofill	Integer	6			
Country	Address.country	0..1	Autofill	Varchar	50			
Marital Status	Patient.maritalStatus	0..1	Optional	Code		http://hl7.org/fhir/R4/valueset-marital-status.html		
Next of KIN Details								
Contact Person's Relationship with patient	Patient.contact.relationship	0..1	Optional	Code		http://hl7.org/fhir/R4/valueset-patient-contactrelationship.html		

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Contact Person Name	Patient.contact.name	0..1	Optional	Varchar				
Contact Person's Mobile Number	Patient.contact.telecom Reference Resource- http://hl7.org/fhir/R4/datatatypes-definitions.html#ContactPoint	0..1	Optional	String				
Hospitalization Details								
Admission Date & Time	Encounter.period		Autofill	DDMMYY YYY HH:MM:SS			These details can be autofilled if the Hospital has an EMR for inpatients	A discharge summary should have Admission, Discharge and transfer details and referral details if there was an interdepartmental, speciality or external referral post admission.
Specialty or Department			Autofill	Integer	2	CD05.090- Specialty list from MDDS which has around 40 specialty listed specific to India.		
Ward Number			Autofill	Integer	2			
Bed No.			Autofill	Varchar	99			
IPD Number			Autofill					
MLC indicator			Mandatory	Integer 1-Yes, 0-No	1			
FIR no.			Conditional	Varchar	50			
Discharge Type			Mandatory	Codeable Concept		CD05.070 01 Cured 02 Expired 03 Transferred 04 DAMA (Discharge Against Medical Advice) 05 LAMA (Leave Against Medical Advice) 06 Absconded 07 Referral	Discharge on clinical advice, patient requested	A discharge summary should have Admission, Discharge and transfer details and referral details if there was an interdepartmental, speciality or external referral post admission.

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Discharge Date & Time	Encounter.period	Optional	Autofill	String				
Discharge Disposition	Encounter.hospitalization.dischargeDisposition	Optional	Mandatory	Integer	2	https://github.com/accesshdi/Health-Data-Dictionary-HDD-for-India/blob/master/JSON/CD05.016_discharge_status_values.json		
If Death, Date of Death	Patient.deceased[x]	Mandatory	Conditional	DDMMYY	8			
Time of Death	Patient.deceased[x]	Optional	Conditional	HH:MM:SS	8			
Primary Cause of Death	Encounter.diagnosis.condition	Mandatory	Mandatory	Varchar	99	ICD 10 code list, Code Directory		
Secondary Cause of Death	Encounter.diagnosis.condition	Mandatory	Optional	Varchar	99	ICD 10 code list, Code Directory		
Clinical Summary								
Patient's Medical History Section								
Main Resource - https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section:MedicalHistory Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-Condition-definitions.html#Condition.code								
Past Health Condition Code (Diagnosis)	Condition.code	0..1	Optional	Varchar	10	ICD10/ SNOMed	Only last diagnosis	
Past Health Condition Onset Date	Condition.onset[x]	0..1	Optional	DDMMYYY	10	SNOMed/ICD		
Past Health Condition end date	Condition.abatement[x]	0..1	Optional	DDMMYYY	10	SNOMed/ICD		
Relevant Clinical Document ID or attachments	Condition.recorder	0..1	Optional	Varchar	20			
Procedure Category	Procedure.category	0..1	Optional	Code		http://hl7.org/fhir/R4/valueset-procedure-category.html		
Procedure Code	Procedure.code	1..1	Mandatory	Code		SNOMed/ICD-PCS	Example- http://hl7.org/fhir	

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
							/R4/valueset-procedure-code.html	
Patient Allergies Section Main Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section:Allergies Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-AllergyIntolerance.html								
Allergy Product Code	AllergyIntolerance.code	1..1	Autofill	Code	5	01 Food 02 Environmental Agent 03 Medicine 04 Animal 05 Pollen 06 Plant 07 Metal 08 Fabric 09 Other Product		
Allergy Substance								
Allergy Reaction Type	AllergyIntolerance.type	0..1	Optional	Varchar		http://hl7.org/fhir/R4/valueset-allergy-intolerance-type.html		
Severity	AllergyIntolerance.criticality	0..1	Optional	Code		01 Mild 02 Moderate 03 Severe 04 Potentially Life Threatening 05 Anaphylaxis 06 Fatal 07 Toxicosis	NRCeS FHIR specification recommends the following value sets- http://hl7.org/fhir/R4/valueset-allergy-intolerance-criticality.html	Recommended codes are provided under column G as these describes clinical criticality of an allergic reaction accurately that an action can be taken on the patient effectively.
Reaction Description	AllergyIntolerance.reaction.manifestation	1..1	Autofill	Code		http://hl7.org/fhir/R4/valueset-clinical-findings.html		
Chief Complaints Section Main Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section:ChiefComplaints Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-Condition-definitions.html#Condition Condition.code								
Chief Complaint Code	Composition.section:ChiefComplaints.code.codi	1..1	Optional	Code		SNOMeD code for presenting complaint		

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
	ng.code Reference Resource-							
Chief Complaint Description	Composition.section:ChiefComplaints.code.text Condition.code.text	1..1	Mandatory	String				
Chief Complaint Duration			Mandatory				Suffering since X days/Months etc	
Physical Examination Section Main Resource- https://www.nrce.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section:PhysicalExamination Reference Resource: Observation- https://www.nrce.in/ndhm/fhir/r4/StructureDefinition-Observation.html								
Examination Type	Observation.method	0..1	Mandatory			01 Inspection 02 Palpation 03 Percussion 04 Auscultation https://github.com/accesshdi/Health-Data-Dictionary-HDD-for-India/blob/master/JSON/CD05.061_examination_type_code_directory_values.json	Physical examination done by physician such as Inspection, Palpation, Percussion, Auscultation	
Examined System	Observation.bodySite	0..1	Mandatory	Codeable Concept		01 Nervous System 02 Eye and Ocular Adnexa 03 Ear and Mastoid (process) 04 Orocraniofacial Region 05 Respiratory System 06 Cardiovascular System 07 Lymphatic System 08 Digestive and Hepatobiliary Tracts and Other Sites within the Abdominal Cavity Not Elsewhere Classified. 09 Genitourinary System 10 Musculoskeletal System 11 Skin, Subcutaneous Tissue and Breast		

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
						12 Body System Not Elsewhere Classified		
Examination Finding (Code If using a terminology standard)	Observation.code	1..1	Mandatory	Code		Example- http://hl7.org/fhir/R4/valueset-observation-codes.html	Physical examination done by physician such as Inspection, Palpation, Percussion, Auscultation	
Examination Finding (Text)	Observation.text	0..1	Mandatory	Text			Narrative	
Observation Value	Observation.value[x]	1..1	Mandatory	Varchar			The information determined as a result of making the observation, if the information has a simple value.	
Diagnosis Section (Active Conditions including comorbidities) Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-Condition.html								
Present Health Condition Onset Date	Condition.onset[x]	0..1	Optional	Date (DDMM YYYY)	10			

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Diagnosis Code	Condition.code	1..1	Mandatory	Varchar	10	ICD10/SNOMeD	Diagnosis for which the treatment was given	
Diagnosis Priority	Condition.severity	Optional	Mandatory	Integer	1	1 - Primary 2 - Secondary		
Investigation Section Main Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section:Investigations Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DiagnosticReportImaging-definitions.html#DiagnosticReport Ref Resource- https://www.hl7.org/fhir/servicerequest.html								
Lab Investigations & Results Main Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section:Investigations Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DiagnosticReportImaging-definitions.html#DiagnosticReport								
Lab Investigation Code	ServiceRequest.code	0..1	Autofill			LOINC for labs		
Lab Investigation Description			Mandatory					
Lab Result Date	Procedure.encounter	0..1	Optional	Date (DDMM YYYY)				
Lab Investigation Result Value	DiagnosticReport.category	1..1	Autofill		10	http://hl7.org/fhir/R4/valueset-diagnostic-service-sections.html		
Lab report Document ID/attachment	DiagnosticReport.code	1..1	Optional					
Result Interpretation	DiagnosticReport.conclusion	1..1	Mandatory					
Radiology Investigations/Procedures & Results Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DiagnosticReportImaging.html								
Radiology Investigation Code	ServiceRequest.code	0..1	Autofill					
Radiology Investigation Description		0..1	Mandatory					

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Radiology Diagnostic Procedure Code	Procedure.code	0..1	Conditional			LOINC		
Radiology Procedure Date	Procedure.encounter	0..1	Conditional	Date (DDMM YYYY)				
Media ID	DiagnosticReport.media.id	0..1	Optional					
Media Link	DiagnosticReport.media.link Reference- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-Media.html	1..1	Mandatory					
Result Interpretation	DiagnosticReport.conclusion	1..1	Mandatory					
Procedure Section Main Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section.Procedures Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-Procedure.html								
Procedure Category	Procedure.category	0..1	Optional	Code				
Procedure Code	Procedure.code	1..1	Autofill	Varchar	10	SNOMeD		
Procedure Date	Procedure.performed[x]	0..1	Mandatory	Date	10			The date element needs extension as it doesn't support Indian Date Format i.e. DD/MM/YYYY
Procedure Site	Procedure.bodySite	0..1	Optional			CD05.026- https://github.com/accesshdigital/Health-Data-Dictionary-HDD-for-India/blob/master/JSON/CD05.026_body_site_master_values.json	Amber	Value sets specified in column G are recommended to be utilized as Indian Healthcare sector is not mature enough to handle complexities and learning curve around SNOMeD value sets provided by FHIR
Procedure Outcome	Procedure.outcome	0..1	Mandatory			SNOMeD	Amber	

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Procedure Notes (Document ID)	Procedure.report Reference (Composition DocumentReference DiagnosticReportLab DiagnosticReportImagin g)	0..*	Optional	Varchar	10			
Discharge Advice								
Medication Orders								
Main Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord-definitions.html#Composition.section:Medications Reference Resource- https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-MedicationRequest-definitions.html#MedicationRequest								
Generic Drug Code	MedicationRequest.medicat ion[x]	0..1	Mandat ory	Integer	5	National Formulary of India/SNOMeD Extension byNRCes	Extension for these fields is required in the NRCes specification as are critical to create a uniform medication order/request	The recommended list medication product list with code provided by FHIR is insufficient for drug prescriptions in India. Hence it is recommended to use either National Formulary of India or generic drugs from NRCes SNOMeD Drug Extension.
Brand Drug Code	Medication.code	0..1	Optional	Integer	10	CD05.105 Amber		The specification should specify that this element will store brand drug code
Strength Value	Medication.ingredient.s trength	1..1	Mandat ory	Varchar	25	Recheck		This field is not covered by the NRCes specifications but is part of the vanilla FHIR resource. NRCes has clubbed it under dosage which can be improved if we have structured way of capturing medication strength, as cost of the databases and ePharmacy has listed the drugs with their strengths.

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber - Element or associated value set needs extension Blue - Additional sections that need to be considered Green - All accepted as it is.
Medication Frequency	MedicationRequest.dosageInstruction MedicationRequest.dosageInstruction.text	1..1	Mandatory	Varchar	60	MDDS Code directory- CD05.023 OD Once a day BD Twice a day TID Thrice a day QID Four times a day AC Before meals PC After meals SOS As required QH Hourly Q2H Every 2 hours HS Before Sleep Q4H Every 4 hours STAT Now		It is recommended to create an extension to use the values provided under CD05.023 for dosage instruction and not keep it free text - https://github.com/accesshddigital/Health-Data-Dictionary-HDD-for-India/blob/master/JSON/CD05.023_medication_frequency_values.json
Route of Administration	MedicationRequest.dosageInstruction.route	0..1	Optional	Varchar	6	https://www.ncres.in/ndhm/fhir/r4/ValueSet-ndhm-route-of-administration.html		
Dose	MedicationRequest.dosageInstruction.doseAndRate.dose[x]	1...1	Mandatory				The amount of the product to be given. This may be a known, measurable unit (e.g., millilitres), an administration unit (e.g. tablet), or an amount of active ingredient (e.g., 250 mg).	
Medication Fills	MedicationRequest.dispenseRequest.number OfRepeatsAllowed	0..1	Optional	Integer	3			
Patient Instructions			Optional	Free Text				
Referral Details								

Data Element (Human Visible Label)	Technical FHIR label specification	Cardinality	Field type	Data Format	Maximum Size	Applicable value set	Remark	Feedback Amber- Element or associated value set needs extension Blue- Additional sections that need to be considered Green- All accepted as it is.
Referral category		0..1	Optional	Char	1	CD05.067 https://github.com/accessdigital/Health-Data-Dictionary-HDD-for-India/blob/master/JSON/CD05.067_referral_category_code_directory_values.json		
Referral From Facility Identification Number		0..1	Mandatory	Integer	2			
Referring Physician ID		0..1	Mandatory	Varchar	10			
Referral document ID		0..1	Optional	Varchar	50		It can be an e-encounter note or e-referral Object	
Referral Facility Identification Number		0..1	Mandatory	Integer	10	Facility Registry		Additional recommended section for discharge summary
Reason for Referral		0..1	Mandatory	Varchat	254			
Referred to Primary Care provider, Doctor's ID		Optional	Conditional					
Author Details								
Author Date	Composition.date	Mandatory		G00.01	10			
Author Time	Composition.attester.time	Optional		HH:MM:SS	8			
Author Identification Number/ID	Composition.attester.party	Optional		Varchar	18			
Author's Digital Signature	Signature.who	Mandatory						

eClaim eObject

An **eClaim** object contains information about a request for reimbursement (“provider payment”) for either of inpatient stay or an outpatient encounter from the provider rendering the services to an appropriate payer who covers this beneficiary.

IRDA codes	Data Element	Data Format	Maximum Size	Field Type	Value Sets/ Code directories MDDS	Values	HDD/Standards	Source	MDDS codes	Remark	Field Category	FHIR resource	Comments
ClaimHeader FHIR resource Type= "Claim Resource" Resource Identifier use= "Claim" Type= "New"or "Resubmitted"													
	Claim Status			System response		active, cancelled, draft, entered in error				Free Text entry	e-object	claim.status	The standard FHIR resource status field value set shall be extended to include the status values as specified in column G
	Claim Date	dd/mm/yy	10	Current date of the system				NHCP	G00.01		e-object	claim.created	
	Claim ID	Varchar	99	System-generated			NHCP	NHCP	05.006.0026		e-object	claim.insurance.identifier	Claim Identifier
	Claim Class	Char	99	Mandatory		1.Cashless 2.Reimbursement	NHCP	HDD			e-object	claim.supportingInfo.value[x].valueString	A new code "ClaimClass" will be created (as key in key value pair array) in claim information category code value set and claim.supportinginfo.category will be "ClaimClass"

IRDA codes	Data Element	Data Format	Maximum Size	Field Type	Value Sets/ Code directories MDDS	Values	HDD/Standards	Source	MDDS codes	Remark	Field Category	FHIR resource	Comments
	Pre-Authorization ID/No.	Varchar	50	Mandatory			MDDS	NHCP	05.006.0030	This number is generated when a new pre-auth is created and will autopopulate from NHCP claim database	e-object	claim.insurance.preAuthRef	
	Payer ID/code	Integer	3	Mandatory	CD05.119 need to be updated by MDDS		MDDS	Payer Registry	05.007.0013	We must update this as public health insurance is not covered	e-object	claim.insurer.identifier	
	Insurance Policy ID/ Health Card ID)	Varchar	50	Mandatory			NHCP	Payer's Product Database	05.006.0006	Should autofill on entering beneficiary ID	e-object	claim.insurance.coverage.contract.identifier	
	Beneficiary ID	Varchar	50	Mandatory			NHCP	Payer's beneficiary Database		Free Text entry	e-object	claim.insurance.coverage.subscriberId	In context of NDHB Report, Beneficiary ID and Patient Health ID may become (coming from Beneficiary ID) then we will consider FHIR label - >claim.patient.identifier to be mapped to Beneficiary ID
	Patient's UHID/ Alternate ID (Till UHID is implemented)	Varchar	50	Mandatory			MDDS		05.002.0002		e-object	Patient.identifier	
	Unique Facility Identification Number	Integer	10	Mandatory	CD05.001-need to be updated		MDDS	Facility Registry-ROHINI 2.0	05.008.0001		e-object	claim.provider.identifier	provider reference =organization resource
	e-Discharge summary	Varchar	15	Mandatory				HIS/PHR			e-object	claim.SupportingInfo.value[5]	A new code "DischargeSummaryURL"

IRDA codes	Data Element	Data Format	Maximum Size	Field Type	Value Sets/ Code directories MDDS	Values	HDD/Standards	Source	MDDS codes	Remark	Field Category	FHIR resource	Comments
	(Document No.)												will be created (as key in key value pair array) in claim information category code value set and claim.supportinginfo.category will be "DischargeSummaryURL"
	Treating/primary Physician ID	Integer	20	Auto filled				HIS/Doctor's Registry			e-object	claim.enterer.identifier	enterer reference =practitioner resource here
	MLC Indicator	Integer	1	Mandatory check		0- No 1- Yes	MDDS	HIS	05.013.0005			claim.supportingInfo.value[x].valueString	A new code "MLIndicator" will be created (as key in key value pair array) in claim information category code value set and claim.supportinginfo.category will be "MLIndicator"
	Scanned PDF Document Attachments if any			Optional								claim.Supporting.value[x].valueAttachment	valueattachment (Base64 blob of documents, or document sFTP url reference) A new code {DocumentType} will be created (as key in key value pair array) in claim information category code value set and claim.supportinginfo.category will be {DocumentType}
Pre-Auth information (retrieved from pre-auth request)													
	Pre-Authorization ID/No.	Varchar	50	Mandatory			MDDS	NHCP	05.006.0030	this data will come from pre-auth	e-object	claim.insurance.preAuthRef	

IRDA codes	Data Element	Data Format	Maximum Size	Field Type	Value Sets/ Code directories	Values	HDD/Standards	Source	MDDS codes	Remark	Field Category	FHIR resource	Comments
	Pre-Authorization Status	Varchar	15	Auto filled		Approved Queried rejected	MDDS Claims Code Directory	NHCP	05.006.003 0	object that was submitted by the provider for this claim	e-object	claim.related.claim.status	
Hospitalization Details (Hospital information system or e-discharge summary)													
	In-Patient no.	Varchar	20	Auto filled			MDDS	HIS or e-discharge summary			e-object	claim.item.encounter.identifier	
	Admission Date	dd/mm/yy	10				MDDS		05.012.000 3		e-object	claim.item.Encounter.Period.start	period datatype is datetime type in FHIR
	Admission Time	HH:MM:SS	8				MDDS		05.012.000 4		e-object	claim.item.Encounter.Period.start	period datatype is datetime type in FHIR
	Utilized Primary (Package / Treatment plan/procedure) Code	Integer	5				MDDS	HIS, pre-auth or e-discharge summary	05.007.003 8		e-object	claim.diagnosis.packageCode	
	Additional Package code if any	Integer	5				MDDS		05.007.003 8		e-object	claim.diagnosis.packageCode	Diagnosis has cardinality as zero to many
	Discharge Date	dd/mm/yy	10	Auto filled			MDDS		05.012.003 8		e-object	claim.item.Encounter.Period.end	
	Primary Physician ID	Integer	20	Auto filled			Doctor's Registry					claim.provider.providerrole.practitioner.identifier	here claim.provider.providerrole.code="primary physician" value to be extended in provider role value set
Minimal Clinical Summary													
	Diagnosis code	Varchar	10	Auto filled			ICD 10	HIS or e-discharge summary			e-object	claim.diagnosis.diagnosis[x].diagnosisCodeableConcept	

IRDA codes	Data Element	Data Format	Maximum Size	Field Type	Value Sets/ Code directories MDDS	Values	HDD/Standards	Source	MDDS codes	Remark	Field Category	FHIR resource	Comments
	Diagnosis Type	Varchar	15		Admitting Diagnosis Differential diagnosis Primary Diagnosis Secondary Diagnosis Discharge Diagnosis		ICD 10	HIS or e-discharge summary			e-object	claim.diagnosis.type	
	Diagnosis Date	Date	8				MDDS	HIS or e-discharge summary			e-object	claim.diagnosis.diagnosis[x].diagnosisreference.condition.recordedDate	
Claim Amount Details													
	Claim Bill Type	Integer	1			1- Hospitalization Bill 2- Pre-hospitalization Bill 3- Post Hospitalization Bill 4- Pharmacy Bill 5- Others 6- OP Bill			05.006.0027		require discussion	claim.item.category	
	Claim Bill ID	Varchar	50	Drop down / Mandatory			MDDS	HIS	05.006.0026		e-object	claim.supportingInfo.value[x].valueString	A new code "ClaimBillID" will be created (as key in key value pair array) in claim information category code value set and claim.supportinginfo.category will be "ClaimBillID"
900000	Package Charges	Decimal	10	Auto filled					05.007.0038			claim.item.unitprice	

IRDA codes	Data Element	Data Format	Maximum Size	Field Type	Value Sets/ Code directories MDDS	Values	HDD/Standards	Source	MDDS codes	Remark	Field Category	FHIR resource	Comments
ICD-PCS	Package Code	Integer	5	Auto filled					05.007.0038			claim.item.productOrService	
	Package Price	Decimal	10	Auto filled					05.007.0021			claim.item.net	
	Primary package unit	Integer	3	Auto filled					05.007.0022			claim.item.quantity	
100000	Room & Nursing charges	Integer	10	Auto filled					05.006.0020	Claim expense type		Category will be mapped to claim.item.category="Room & Nursing" Charges will be mapped to claim.item.net	
200000	ICU charges	Integer	10	Auto filled					05.006.0020	Claim expense type		Category will be mapped to claim.item.category="ICU" Charges will be mapped to claim.item.net	
600000	Investigation Charges	Integer	10	Auto filled					05.006.0020	Claim expense type		Category will be mapped to claim.item.category="Investigation" Charges will be mapped to claim.item.net	
300000	OT charges	Integer	10	Auto filled					05.006.0020	Claim expense type		Category will be mapped to claim.item.category="OT" Charges will be mapped to claim.item.net	
400000	Medicine & Consumables charges	Integer	10	Auto filled					05.006.0020	Claim expense type		Category will be mapped to claim.item.category="Medicine & Consumables" Charges will be mapped to claim.item.net	
500000	Professional fees charges	Integer	10	Auto filled					05.006.0020	Claim expense type		Category will be mapped to claim.item.category="Professional Fee" Charges will be mapped to claim.item.net	
700000	Ambulance charges	Integer	10	Auto filled					05.006.0020	Claim expense type		Category will be mapped to claim.item.category="Ambulance"	

IRDA codes	Data Element	Data Format	Maximum Size	Field Type	Value Sets/ Code directories MDDS	Values	HDD/S tand ar ds	Source	MDDS codes	Remark	Field Categor y	FHIR resource	Comments
												Charges will be mapped to claim.item.net	
Claim Payment Summary													
	Total Claim Amount	Decimal	10	Auto filled					05.006.002 1			claim.total	
	Copayment /patient payable amount if any	Decimal	10	Auto filled			MDDS		05.007.003 2			claim.supportingInfo.value[x].valueString	A new code "CoPayment/PatientPayable" will be created (as key in key value pair array) in claim information category code value set and claim.supportinginfo.category will be "ClaimBillID"
	Net Payable	Decimal	10	Auto filled			MDDS					claim.supportingInfo.value[x].valueString	A new code "NetPayable" will be created (as key in key value pair array) in claim information category code value set and claim.supportinginfo.category will be "ClaimBillID"

Drug Index Specifications

MDDS Entity	MDDS Code Directory	Field Code	Field Name	Cardinality	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
40									New Entity in MDDS

MDDS Entity	MDDS Code Directory	Field Code	Field Name	Cardinality	Data Format String, Varchar, Integer, Value set	Maximum Size	Applicable code directory/value set	MDDS Label	Remark
	CD05.142 or CD05.104								New or existing Code Directory in MDDS
			Generic Drug Code	1	Varchar	10	CD05.104 or CD05.142		
			Generic Drug classification (Analgesic, antibiotics, etc.)	1-*	Varchar	10	CD05.106 or CD05.142		Existing MDDS or International Std ATC classification
			Generic Name	1	Varchar	10	CD05.104 or CD05.142		
			Active ingredients	1 - *	Varchar	100	CD05.143		New Code Directory to be created
			Brand Name	1	Varchar	100			To be mapped from Multiple Data Sources
			Manufacturer Name	1	Varchar	100			To be mapped from Multiple Data Sources
			Manufacturer License details	1	Binary				To be uploaded as a digital document or certificate
			Drug Form (liquid, powder, etc.)	1-*	Varchar	100	CD05.108		
			Drug-Drug Interactions	0-*	Varchar	500			To be linked to an interaction database
			Dosage	1-*	Varchar	500			
			Pharmaceutical Unit of Measurement	1-*	Varchar	20	CD05.109		
			Route	1-*	Varchar	100	CD05.111		
			Alternate Drug	1 - *	Varchar	100	National Drug Index Code Directory		
			Indications	1-*	Varchar	500			Link to DGCI/CDSCO databases
			Contraindications	1-*	Varchar	500			Link to DGCI/CDSCO databases

References

References

- + Health spending pushed 5.5 Crore into Poverty <https://economictimes.indiatimes.com/news/politics-and-nation/health-spending-pushed-55-million-into-poverty-in-a-year-study/articleshow/64568199.cms>
- + Drug Costs and Poverty https://cddep.org/blog/posts/63_million_indians_are_pushed_poverty_health_expenses_each_year-and_drugs_are_chief_cause/
- + Metadata and Data Standards for Health Domain Part I - <https://main.mohfw.gov.in/sites/default/files/Part-I%20Overview%20Report%20Health%20MDDS.pdf>
- + National Committee on Metadata and Data Standards for Health Domain Part II: http://nhsrcindia.org/sites/default/files/Part-II%20Data%20Element%20Quick%20Ref%20Health%20MDDS_0.pdf
- + National Committee on Metadata and Data Standards for Health Domain Part III: http://egovstandards.gov.in/sites/default/files/Part%20III%20Code%20Directory_0.pdf
- + No. of Mobile Phone users – 81.32 crores <https://www.statista.com/statistics/274658/forecast-of-mobile-phone-users-in-india/>
- + DBT Beneficiaries – 77.7 crores <https://dbtbharat.gov.in/>
- + No of Aadhar cards issued 124.3 crores <https://uidai.gov.in/>
- + UPI Crossing 100 Million Users / UPI has crossed 1 Billion transactions https://m.timesofindia.com/business/india-business/upi-hits-1-billion-transactions-in-october-plans-to-go-global/amp_articleshow/71787729.cms
- + No. of PMJAY Cards issued 100,015,885 <https://pmjay.gov.in>
- + Provider Working Groups Concept Note: <https://drive.google.com/file/d/1zwCc4YMJH9c6Dh7XVQga7DXNNemZ5odT/view?usp=sharing>
- + <https://ndhm.gov.in/>
- + <https://ndhm.gov.in/home/ndhm>
- + <http://www.experian.com/blogs/healthcare/2019/10/top-5-benefits-healthcare-interopability/#:~:text=Interoperability%20currently%20remains%20a%20challenge,ultimately%20improve%20the%20patient%20experience.>

- ✚ Presentation on what are eObjects and How to use them: <https://drive.google.com/open?id=1i48bAib4lAdbeZqsRm2IsytdbFrYaCXk>
- ✚ eObjects v1 Details: <https://drive.google.com/open?id=1p622-1yaeTdu3dkHKYbHckU29XsX2R-c>
- ✚ eObjects v2 Details: <https://drive.google.com/file/d/1a57duT6AKgZLP5bqhTYy6xb9e-Wo1Eib/view?usp=sharing>
- ✚ eDischarge Summary v3 Including the recommendations from NRCeS:
https://drive.google.com/file/d/1scX1xXzOl9BgZls6GuK01Wjj_gwwnQwD/view?usp=sharing
- ✚ Code Directory Details: https://drive.google.com/file/d/1lzn_0a6fb6Z9qhbcbgHHH2o8kjjZmk67D/view?usp=sharing
- ✚ Sample JSON Format eObject: <https://drive.google.com/file/d/1MKOkOIsjdIJHyTSS-FcKCGy0jXeyHZEh/view?usp=sharing>
- ✚ <https://timesofindia.indiatimes.com/blogs/voices/telemedicine-the-next-technology-frontier-in-the-healthcare-ecosystem-in-india/>
- ✚ <https://inc42.com/datalab/telemedicine-market-opportunity-in-indian-healthtech/>
- ✚ <https://www.liebertpub.com/doi/10.1089/big.2013.0023>
- ✚ Diversity of India's Healthcare Provider Sector is as per Strategy Council, ACCESS Health internal research
- ✚ Fuzzy boundaries is as per Strategy Council, ACCESS Health internal research.